Tone diachrony beyond tonogenesis: Reconstructing the tone melodies of nouns in Proto-Edoid (Niger-Congo)

1 Introduction[1]

In the diachronic tone literature, original tonogenesis events involving the ‘transphonologization’ of a segmental contrast to a tonal one are widely studied (Kirby, Pittayaporn, & Brunelle 2022 – see within for many references). Much less focus has been placed on how tone systems evolve after tone is firmly established, and what the implications are for both the study of tone and sound change generally. How far back can tone systems be reconstructed in language families without any clear tonogenesis event? This is a question which hovers over comparative Africanist linguistics, where the great majority of the continent’s languages are tonal with no indication that this is a recent phenomenon. Thus, it is generally assumed that the massive Niger-Congo phylum had tone even in its earliest proto-stage (Hyman 2020) rendering any original tonogenetic event irrecoverable, and that Niger-Congo languages without tone are innovative (e.g. famously in Swahili). In fact, evidence for *any* original tonogenetic event in any African family is remarkably lacking compared to other tonal areas of the world (Sæbø, Grossman, & Moran 2025).[2]

Occasionally, dedicated studies have sought to specifically reconstruct the tone and tone patterns for individual African families using the methodologies of historical linguistics, some notable examples including Greenberg (1948) on Proto-Bantu tone, Moñino (1995) on Proto-

[1] In alphabetical order, I would like to thank …

[*to be included*]

All errors are my own.

[2] That is, evidence for ‘strict tonogenesis’ in Africa is lacking, whereby a non-tonal language acquires a tonal contrast for the first time (Sæbø, Grossman, & Moran 2025; see also Hyslop 2022).

Gbaya, Boyeldieu (2000) on Proto-Sara-Bongo-Bagirmi, Elderkin (2008) on Proto-Khoe, *inter alia*. Still, tone in the vast majority of families remains without a rigorous application of the comparative method. This paper represents one such attempt, with the immediate goal of reconstructing the tone patterns of nouns in the Edoid family, one branch within Benue-Congo, which itself is the largest branch within Niger-Congo (the largest phylum in Africa). To this end, we have created the largest lexical database to date on Edoid languages as part of the 'Proto-Edoid Tone Project', or 'PEdT Project' for short. The PEdT database consists of 3,038 individual data points in 21 modern Edoid languages drawn from 38 sources (included in its entirety in the supplemental materials). Altogether, 436 cognates are identified based on Elugbe's (1989a) sound correspondences and Proto-Edoid reconstructions. We draw strictly from nouns, as verbs do not contrast for lexical tone.

Based on this database, we systematically compared tone patterns across the four proposed branches of Edoid based on the aforementioned Elugbe study: Delta Edoid (DE), South-West Edoid (SW), North-Central Edoid (NC), and North-West Edoid (NW). A number of tonal correspondences within each branch were established, which we reconstructed as combinations of tone on a noun class prefix and on the stem called 'nominal tone melodies' (e.g. NW is reconstructed with two melodies *L-L and *H-L). The melodies reconstructed in each branch were then systematically compared against each other in order to reconstructed Proto-Edoid itself. The result was eight tonal correspondence sets reconstructed with a symmetrical set of melodies, namely **L-L, **L-LH, **L-H, **L-HL, **H-L, **H-LH, **H-HL, and **H-H (two asterisks indicate Proto-Edoid, whereas one is for an intermediate reconstruction at the branch level).

Using these PE melodies as our starting point, we can track several recurring tone changes in Edoid diachrony. Two types involve shifting of H tone, most commonly rightward shift such as what we term the Great Edo Tone Shift whereby the Proto-NC *H-L melody becomes L-H in the language Edo. Leftward shift of H is also hypothesized (i.e. PE **L-H to P-NC **H-L) but this is less common, in line with the tendency for tone effects to be perseverative rather than anticipatory (Hyman 2017). Other recurrent changes include H-Obligatoriness (i.e. adding a H tone to an all-low word) and H-Lowering (i.e. H becoming L, M, or downstepped H, whether conditioned or unconditioned). These tone changes show the hallmarks of regular sound change, and thus demonstrate the success of applying the comparative method to tone melodies (though unexpected tonal outcomes for individual cognates are not uncommon and require further scrutiny).

As is common with most comparative work in Africa the data is skewed, with large empirical gaps in the database (especially for the SW and NW branches). Our correspondence sets and reconstructions make predictions as to which data we should find with further data collection, which in turn will refine this initial proposal. By grouping Proto-Edoid into these eight tone melodies, we can also begin to systematically compare tone here to other Benue-Congo branches, such as Proto-Yoruboid, Proto-Igboid, *etc*. Preliminary comparison within Benue-Congo does not yet reveal any clear tonal correspondences, and we conclude that this step must be faithfully completed before we can entertain the tone system of higher-order Proto-Niger-Congo.

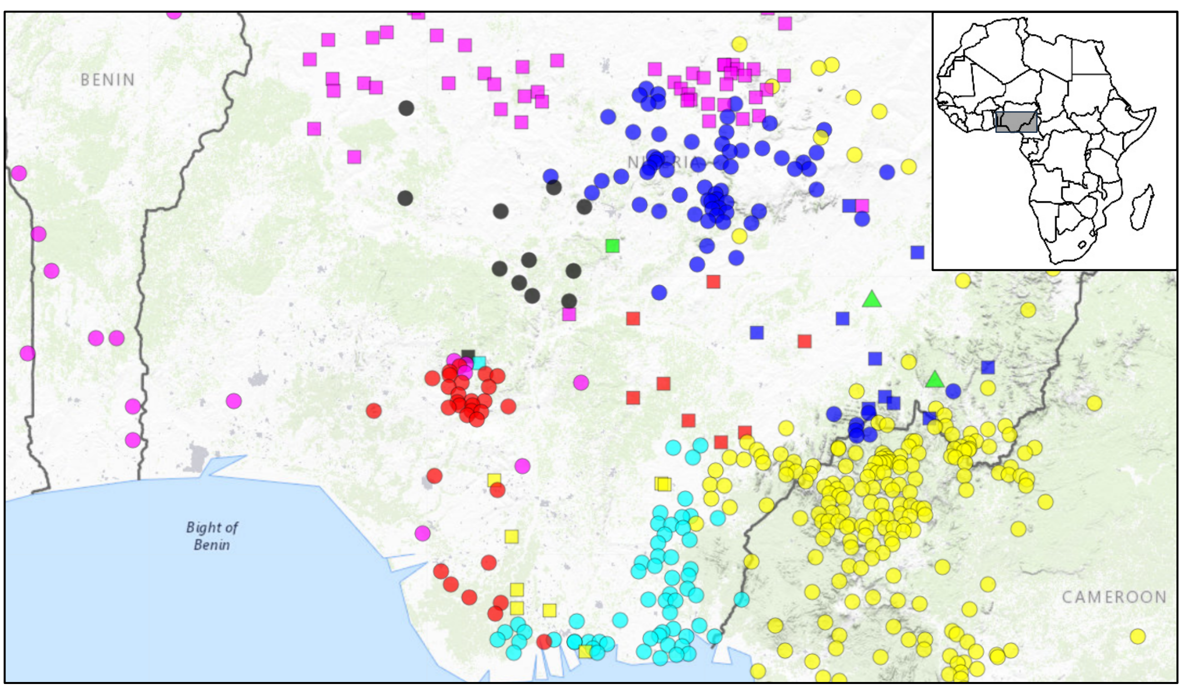
Following this introduction (§1), this paper is organized as follows. The relevant background on Edoid is provided in §2, followed by the data and methods of the PEdT Project in §3. Next, §4 provides the reconstructed tone melodies within the four Edoid branches, followed by a reconstructed of tone in Proto-Edoid itself in §5. Then §6 provides discussion, centered around common tone changes, the regularity of tone change and innovation vs. retention of tone melodies, and prospects for reconstructed Proto-Benue-Congo and Proto-Niger-Congo. Finally, §7 provides a brief summary. Two appendices are provided, one repeating the consonantal correspondences of

Elugbe (1989a) which form the primary basis for cognacy, and the other appendix grouping the 436 cognates into the eight reconstructed proto-melodies.

2 The Edoid family

The Edoid family is a subgroup within Benue-Congo, itself one of the main subgroupings within the transcontinental Niger-Congo phylum (Elugbe 1989c). Edoid is spoken entirely within southern Nigeria, bordered largely by other Benue-Congo families, most prominently the Yoruboid complex to the west and the Igboid complex to the east. The Edoid family constitutes approximately 25 to 30 languages (29 are listed on Glottolog – Hammarström *et al.* 2024), with substantial dialect diversity. The map in (1) locates Edoid within Nigeria (the red circles ●), mapped against the other major branches of Benue-Congo (taken from Glottolog). We remain neutral as to the internal subgrouping of the major Benue-Congo branches and follow the rake-like structure of Williamson (1989:261) (also assumed by Glottolog).

(1) Major Benue-Congo branches: Edoid ●, Yoruboid ●, Igboid ■, Delta Cross ●, Bantoid ● (which includes Bantu), Idomoid ■, Jukunoid ■, Nupoid ●, Platoid ●, Kainji ■



Elugbe (1973, 1989a) systematically identifies the segmental correspondences of Edoid languages, reconstructing approximately 200 Proto-Edoid words grouped into four branches based on regular sound changes and lexical innovations. The branches are geographically-labeled: North-West Edoid (NW), North-Central Edoid (NC), South-West Edoid (SW), and Delta Edoid (DE) located furthest south (in the Niger River Delta). We examine 21 individual Edoid languages in this study, listed in (2) and split up based on their membership in one of the four Edoid branches. We include both the ISO code and the glottocode for each language for referential purposes. Hereafter we shall exclusively use the ISO codes, both for their brevity and their recognizability. The approximate geographic location of these 21 languages is mapped in (3), which includes several such ISO codes.

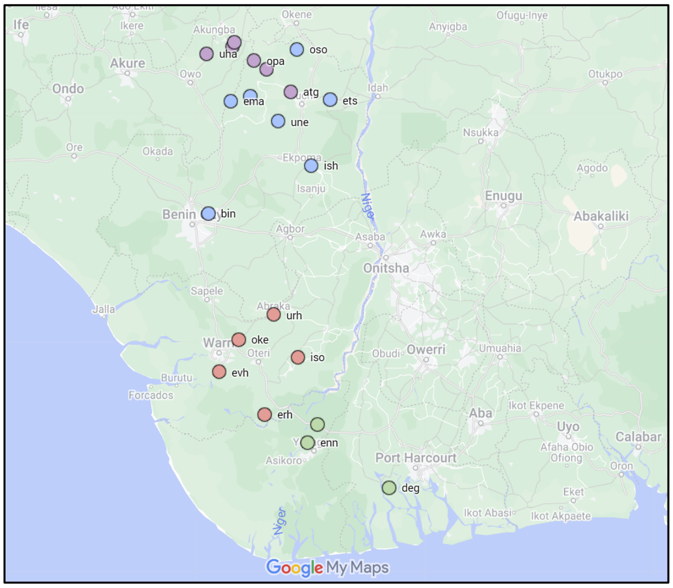
(2) Four branches of Edoid and the languages examined (*n*=21)[3]

	Branch	Language	ISO	Glottocode	Coordinates	
a.	DE	Degema	[deg]	dege1246	4.73	6.81
	DE	Engenni	[enn]	enge1239	5.01	6.31
	DE	Epie-Atissa	[epi]	epie1238	5.12	6.37
b.	SW	Eruwa	[erh]	eruw1238	5.18	6.05
	SW	Isoko	[iso]	isok1239	5.53	6.25
	SW	Okpe	[oke]	okpe1250	5.64	5.89
	SW	Urhobo	[urh]	urho1239	5.79	6.10
	SW	Uvwie	[evh]	uvbi1238	5.44	5.77
c.	NC	Edo	[bin]	bini1246	6.41	5.70
	NC	Esan	[ish]	esan1238	6.70	6.33
	NC	Emai-Iuleha-Ora	[ema]	emai1241	7.09	5.84
	NC	Etsako	[ets]	yekh1238	7.10	6.45
	NC	Uneme	[une]	unem1238	6.97	6.13
	NC	Ghotuo	[aaa]	ghot1243	7.12	5.96
	NC	Ososo	[oso]	osos1238	7.41	6.25
d.	NW	North Ibie	[atg]	ivbi1241	7.15	6.21
	NW	Oloma	[olm]	olom1241	7.28	6.06
	NW	Okpamheri	[opa]	okpa1238	7.34	5.98
	NW	Uhami	[uha]	uham1238	7.38	5.69
	NW	Ehueun	[ehu]	ehue1238	7.45	5.86
	NW	Ukue	[uku]	ukue1238	7.42	5.85

[3] The classification of Ososo [oso] as North-Central Edoid and North Ibie [atg] as North-West Edoid comes from Lewis (2013). These languages were not surveyed in Elugbe's work. The Edoid languages not surveyed in this paper are Ikhin-Arokho [ikh], Enwan [env], Ikpeshi [ikp], Igwe [igw], Sasaru [sxs], Iyayu [iya], Akuku [ayk], and Okpe [okx]. We also do not include individual languages which have at one point or another been classified as sister to the Edoid family, such as Akpes [ibe], Arigidi [aqg], and Ukaan [kcf] (see also §6.4).

(3) Approximate coordinates of Edoid languages of this study, colored by branch

(Purple ● = NW, Blue ● = NC, Red ● = SW, Green ● = DE)



Like most languages of the area, Edoid languages have a largely analytic and head-initial syntax (e.g. verb-object order, noun-modifier order, prepositions), with limited segmental inflection (mostly prefixal noun classes and remnant inflectional suffixes on verbs). Phonologically, there are no codas and consonant clusters only involve approximants. Certain Edoid languages are known for a fortis/lenis distinction in consonants (Ladefoged 1968; Elimelech 1976; Elugbe 1989a), which cannot be straightforwardly reduced to another phonetic dimension (e.g. voicing, gemination, frication, *etc.*). Elugbe (1989a) reconstructs a full set of fortis/lenis contrasts, though this system has since collapsed in most descendent Edoid languages. In Proto-Edoid and in descendent languages which retain the fortis/lenis contrast, the lenis version is transcribed with an <h> (i.e. fortis **íkpò** 'clothes' vs. lenis **íkphò** 'erosion' in Etsako – Elimelech 1976:8). Moreover, many of the languages have active Advanced Tongue Root (ATR) harmony in the vowel system (i.e. [+ATR] **i e ə o u** vs. [-ATR] **ɪ ɛ a ɔ ʊ**) also reconstructed for Proto-Edoid, though in the central core this has broken down considerably (Rolle, Lionnet, & Faytak 2020).

All Edoid languages are robustly tonal, mostly with a high tone (H) versus low tone (L) distinction, but a few languages show evidence for mid tone (M) (e.g. Ghotuo – Elugbe 1973, 1989b). Depending on the language, downstep may occur but typically only affects high tones (i.e. ꜜH). The difference between M and ꜜH depends on their phonological behavior, and the two transcriptions may be found even for the same language across sources (e.g. Urhobo). Contour tones on syllables occur but are not particularly frequent.

In the entire Edoid family, only nouns have lexical tone contrasts whereas tone on verbs is strictly grammatical. For example, minimal pairs in Esan exist for nouns (e.g. **áwà** 'dog' vs. **àwà** 'forbidden thing') but none exist for verbs, where tone differences strictly express temporal, aspectual, and polarity-based inflection, in (4). As such, our study only surveys nouns.

(4) Esan – Tone on verbs is exclusively grammatical (author fieldnotes)

a. *Akhere ghé ọgọ*

àxérè ɣé **ɔ̀gɔ́** [**àxérè ɣɔ̂gɔ́**]

Akhere look.at\PERFECTIVE bottle 'Akhere (has) looked at a bottle'

b. *Akhere ghè ọgọ*

àxérè ɣè **ɔ̀gɔ́** [**àxérè ɣɔ̀gɔ́**]

Akhere look.at\IMPERFECTIVE bottle 'Akhere is looking at a bottle'

Edoid nouns canonically consist of a noun class prefix and a nominal stem (most often one to two syllables). Across Edoid, the class prefix is almost always a single vowel, though in a few languages some class prefixes bear an onset. They function primarily to indicate number, with class prefixes forming singular/plural pairs, as in (5).

(5) Edoid – Noun class prefixes forming singular/plural pairs

a. Oloma – System intact (Elugbe & Schubert 1976:74)

Singular	**lé-nwè**	‘breast’		**ghɔ́-bì**	‘leaf’
Plural	**í-nwè**	‘breasts’		**ɛ́-bì**	‘leaves’

b. Edo – System broken down (Agheyisi 1990:39,87)

Singular	**ò-kpìá**	‘man’	Cf.	**è-bé**	‘leaf, leaves’
Plural	**ì-kpìá**	‘men’			

Twelve such pairings of class markers are reconstructed for Proto-Edoid (Elugbe 1989a:123-124). In some languages this system is still largely intact (e.g. in Oloma above), while in others it is broken down with only a few nouns showing any morphological alternation (e.g. in Edo).

The noun class system plays a role in the tonal analysis to come, and we shall therefore demarcate the prefix both in transcribing the word and in analyzing its tone pattern (e.g. for **ò-kpìá** above, the tone pattern is rendered L-LH). We make this demarcation even in those words where there is no longer a regular alternation (e.g. **è-bé** above). Interesting, even in languages where the noun class system is all but obsolete, there remains a requirement for all nouns to begin with a vowel, reflected in recent loanwords. For example, in Edo a prothetic **e-** is added to **è-gɔ́mù** ‘gum’, **è-fàdá** ‘Catholic priest(s)’, **è-lɛ́dì** ‘lead pencil(s)’, *etc*. (Agheyisi 1986).

3 Data and methods

The PEdT Project employs the traditional Comparative Method but applied to tone patterns, whereby daughter languages are systematically compared in order to reconstruct the ancestral state of the mother language, ultimately Proto-Edoid (see Dockum 2019 for many references on the Comparative Method, and in particular its application to tone). All currently available data on Edoid was consulted for this paper. Fieldwork on various Edoid languages by Ben O. Elugbe was the starting point (Elugbe 1973, 1977, 1986, 1989a, 1989b, 1989c; Elugbe & Schubert 1976). This seminal work was complemented by an exhaustive survey of the Edoid literature, where lexical data was collected if there appeared to be consistent tonal transcription (Thomas & Williamson 1967; Ladefoged 1968; Mafeni 1969; Welmers 1969; Elimelech 1976; Thomas 1978; Iweh 1983; Aigbe 1985; Eigbedion 1985; Agheyisi 1986; Blanc 1986; Donwa-Ifode 1986, 1989; Ejele 1986; Masagbor 1989; Klomp 1993; Emuekpere-Masagbor 1997; Aziza 1997, 2003, 2008; Osiruemu 2005; Ukere 2005; Schaefer & Egbokhare 2007; Kari 2008; Olaoluwa 2015; Legbeti 2022).[4] Lastly, when possible these data were supplemented by personal fieldnotes (specifically on Degema, Esan, Urhobo, and Okpe).

From these sources, the PEdT database was created. This consists of 436 cognates based on 3,038 individual data points, using only lexical items which are cognate in at least two Edoid languages. Each cognate is provided a unique code for sorting and referential purposes, as well as a cognate name, a mnemonic based on characteristic segments across languages for this cognate (note that this is not a formal segmental reconstruction), its semantic domain, the number of languages with this cognate, and its meaning in daughter languages. A sample cognate is in (6) for SMOKE_1.

(6) Sample cognate entry from the PEdT database

PE_code	PE_cognate	mnemonic	domain	languages	meaning
PE_549	SMOKE_1	*UBHURI*	substance	8	smoke; ashes [oke]

[4] Unfortunately, certain sources could not be included in the database due to transcriptional irregularities. For example, Uvwie words **ètú** ‘cap’ **ógbéɪ̃̀** ‘tortoise’, and **ètè** ‘snake’, *etc*. from Ikoyo-Eweto & Ekiugbo (2017) are transcribed as **ètù** ‘cap’ **ògbéɪ̃̀** ‘tortoise’, **èté** ‘snake’ in later work (in Ekiugbo & Ugorji 2019). It is unclear what to attribute this variation to, and we therefore set such sources aside for this paper.

Following each cognate, descendent forms are provided from the relevant languages. We include the original data point (including its source, in abbreviated form), as well as a normalized segmental and tonal rendition of the cognate (or the relevant portion if in a compound). As stated in the previous section, we universally demarcate the class prefix from the stem. We continue our sample cognate for SMOKE_1 in (7) below, divided into the four Edoid branches.

(7) Sample cognate entry from the PEdT database (continued from (6))

	branch	language	iso	data_point	segments	tone	source
a.	DE	Engenni	[enn]	uḅuru ẹ́tạị	**u-ɓuru**	L-LL	{tho67}
		Epie-Atissa	[epi]	uḅuru táān	**u-ɓuru**	L-LL	{tho67}
b.	SW	Isoko	[iso]	ívbìrrì	**i-ʋiri**	H-LL	{elu77}
		Okpe	[oke]	íìrì̃	**i-irĩ**	H-LL	{don89}
		Urhobo	[urh]	íʋìṛì	**i-ʋiri̥**	H-LL	{azi97}
		Uvwie	[evh]	íʋiṛi	**i-ʋiṛi**	H-LL	{don89}
c.	NC	Ososo	[oso]	ɛ̀wɔ̀	**ɛ-wɔ**	L-L	{ola15}
d.	NW	North Ibie	[atg]	èwọ̀lì	**e-wɔli**	L-LL	{mas89}

Cognacy was based on established consonantal correspondences in the nominal stem, as established previously by Elugbe (1989a) which we reproduce in Appendix 1. For each of the 436 cognates, we provided a confidence rating 'high', 'medium', and 'low'. Those words which complied with these previously established consonantal correspondences were designated with 'high confidence' (n=190). A sample of such cognates is in (8). In the first two – cognates for WATER and ELEPHANT – all languages shared the same general shape. Here, the stem-initial consonant is identical and in red (and the proposed Proto-Edoid form from Elugbe 1989a is also provided at the top, indicated with two **). Other examples in (8) include cognates where there is a systematic divergence from the proto-segment in some language, such as in MARKET with ****k** and CHIN with ****gbh** (recall that the <h> stands for a lenis contrast), but for which Elugbe identified as regular. The bulk of changes involve the development of lenis consonants, as in the final three columns. Here, there are systematic changes within the respective branches which result in regular correspondence (e.g. ****th** becomes DE /**t**/ but SW /**r̥**/ ~ /**r**/).[5]

(8) Cognates of ‘high confidence’ with expected segmental correspondence (*n*=190)

		WATER (PE: **A-mɪN)	ELEPHANT (PE: **E-ni)	MARKET (PE: **A-ki)	CHIN (PE: **A-gbhamhɪ)	PALM_1 (PE: **U-ɗi)	GREY_HAIR (PE: **ghU-daN)	TONGUE (PE: **U-dhamhɪ)	TREE (PE: **U-thaNɪ)	EAR (PE: **ghU-chɔGi)
a.	[deg]	**à-mín**	**è-ní**	**è-kí**	**ʊ̀-ŋgbá**	**ù-ɗí**	**ʊ̀-dáɲ**	**ʊ̀-ɗɛ́m**	**ʊ́-ꜜtáɲ**	**ʊ́-ꜜsɔ́**
	[enn]	**à-mìnì**	-	**è-kì**	**à-gbà**	**ù-ɗì**	**ɛ̀-dàà**	**ɔ̀-ɗɛ̀mʊ̀**	**ɛ́-tàì**	**ɛ́-sɔ̀ʊ̀**
	[epi]	**à-mìnì**	-	**è-kì**	**à-gbȁ**	**ì-ɗì**	**ì-dàn**	**ì-ɗɛ̀mʊ̀**	**ì-táān**	**ì-sɔ́ɔ̄**
b.	[erh]	**à-mí**	-	**è-kí**	**à-gbà**	**è-dí**	-	**ʊ̀-rɔ̀ʊʊ́**	**ʊ́-rȁrɛ̏**	-
	[iso]	**à-mì**	**è-nì**	**è-kì**	**à-gbà**	**ú-dí**	-	**ɛ̀-ɾɔ̀ʊ̀**	**ú-ré**	**ʊ́-zɔ́**
	[oke]	**á-mì**	**è-nĩ́**	**é-jì**	**ɛ-gbȁ**	**ù-dì**	**é-r̥à**	**ò-rɛ́mʊ̀**	**ò-r̥ã́**	**ò-r̥ɔ́**
	[urh]	**à-mè**	**è-nĩ̀**	**è-kì**	**é-gbã̄**	**ù-dì**	**è-ʃà**	**ɛ̀-ɻèβɛ̀**	**ú-r̥é**	**ó-r̥ɔ́**
	[evh]	**à-mì**	**è-nì**	**è-kì**	**à-gbàʊò**	-	-	-	**ʊ̀-rá**	**è-sɔ̃́**
c.	[bin]	**à-mɛ̃̀**	**è-ní**	**ɛ̀-kì**	**à-gbàʊ̃ɛ̃̀**	**ù-dĩ́**	**è-dɛ́**	**á-ɽáʊ̃ɛ̃̀**	**è-r̥ã́**	**è-hɔ́**
	[esa]	**à-mɛ̃̀**	**é-nì**	**è-kì**	**à-gbȁ**	**ú-dĩ̀**	-	**ó-làmè**	**é-rȁ**	**é-hɔ̀**
	[ema]	**à-mɛ̀**	**í-nì**	**è-kĩ̀**	**à-gbȁ**	**é-dĩ̀**	**é-dɛ̀**	**ó-ɛ̃̀mì**	**ó-rȁ**	**é-hɔ̃̀**
	[ets]	**à-mɛ̀**	**í-nì**	**à-kì**	**à-gbà**	**ú-dì**	**é-dɛ̀**	**ó-lɛ̀mhì**	**ó-r̥à**	**é-sɔ̀**
	[une]	**à-mɛ̀**	**í-nì**	**à-kì**	**à-gbà**	**ú-dì**	**é-dɛ̀**	**ɛ́-r̃ɛ̀mhì**	**ó-θà**	**é-sɔ̀**
	[aaa]	**à-mɛ̀**	**ī-ní**	**ghò-kì**	**à-gbà**	**ghí-ɹì**	**ghō-dɛ̀**	**ghí-ɹɛ̀**	**ō-tà**	**ghō-hɔ̀ɣì**
	[oso]	**à-mɛ̀**	-	**ò-xì**	**[í-zù]à-gbà**	-	-	**írɛ̀-r̥ɛ̀**	**ó-thɛ̀**	**ó-sɔ̀**
d.	[atg]	**à-mɛ̀**	-	-	**è-kpàmì**	-	**é-dɛ̀**	-	**ó-rɛ̀**	**é-sɔ̀**
	[olm]	**â-mɛ̀**	**é-nì**	**ghè-kì**	**à-gbà**	**ghú-dì**	**ghó-zɛ̀**	**ɔ́-lɛ̀mhì**	**ɔ́-shà**	**ghɔ́-z̈ɔ̀**
	[opa]	**à-mɛ̀**	-	**wè-kì**	**à-gbà**	-	-	**ʊ́-rɛ̀**	**ʊ́-ʃà**	**wʊ́-z̈ɔ̀**
	[uha]	**à-mɛ̀**	**è-nì**	**è-cì**	**à-gbàmì**	**ì-dì**	-	**ɛ̀-dɔ̀mì**	**ó-ʃà**	**ɛ́-sɔ̀**
	[ehu]	**à-mɛ̀**	-	**è-hì**	**à-wȁ**	-	-	-	**è-ʃā**	**ē-ɹ̥ɔ̀**
	[uku]	**à-mɛ̀**	-	**è-kì**	**à-gbȁ**	**í-dì**	-	**ɛ́-dàmì**	**ò-ʃâ**	**é-ɹ̥ɔ̀**

[5] We based our confidence rating primarily on Elugbe’s consonant reconstructions. Vowels were much more variable, and subject to various harmony processes, and noun class substitution. In general, however, major vocalic features were preserved across cognates (i.e. height, backness, ATR, *etc.*).

There were also cognates rated with 'medium confidence' (n=171). These words deviated from the expected segment correspondence typically with respect to the stem-initial consonant, a sample of which is provided in (9). For example, ANTELOPE_2 has a correspondence between **j** in DE languages and **z** in the other branches which is not otherwise established in Elugbe (1989a) (perhaps a reflex of **ɟ ~ **j variation in Proto-Edoid for this word). We assume cognacy if there is both a shared meaning, and there are additional non-trivial phonological properties in common, e.g. in terms of sharing a major consonantal feature (e.g. place of articulation), containing the same number of syllables, sharing vowel quality, *etc*.

(9) Cognates of ‘medium confidence’ – Non-established but plausible correspondence (*n*=171)

		ANTELOPE_2 (j↔z)	MUSHROOM (s↔t)	CALABASH_4 (gb↔b↔β)	FOOT_LEG_2 (β/v/ʋ↔w↔∅↔mh)	FOOD (m↔ʋ↔b↔n)	COUGH_N (j↔w↔hw↔ŋm↔h)	SNAIL (s↔r̥↔r↔θ↔l↔th)	BLOOD_2 (z↔gj↔dʒ↔d↔dz↔r)
a.	[deg]	**ú-ꜜjów**	**ù-sùɗú**	-	**à-βí**	-	-	-	**ì-zàlà**
	[enn]	**ú-jóù**	-	**à-gbìnà**	**à-vì**	-	-	-	-
	[epi]	-	-	**è-gbèlè**	**à-vì**	-	**ʊ̀-jílɔ̄**	-	**ì-zàrà**
b.	[erh]	-	-	-	**à-ʋɔ̀**	-	-	-	**à-zí**
	[iso]	-	-	**ò-bèlè**	**ʊ̀-ʋɔ̀**	-	-	**ù-sèkpè**	**à-zɛ̀**
	[oke]	-	-	-	**ò-wɔ̀**	**è-márɛ̃́**	-	-	-
	[urh]	**ú-zò**	**ì-tú**	-	**ò-wɔ̀**	**è-màrè**	**ó-wɔ̀r̥è**	**ú-sèkpè**	-
	[evh]	-	-	**è-bèlè**	**à-ʋɔ̀**	-	-	-	-
c.	[bin]	**ú-zò**	**ù-tṹ**	-	**ò-wɛ̀**	**è-ʋàré**	**ó-hwɛ̃́**	**ú-r̥èé**	**è-ságjɛ̃̀**
	[esa]	**ú-zò**	**ú-tũ̀**	-	**ò-ɛ̀**	**é-bàlè**	**ó-ꜜhwɛ̃̂**	**ú-ꜜr̥ê**	-
	[ema]	**ú-ꜜzó**	**í-tùú**	**ù-bèlè**	**ò-ɛ̀**	**é-màè**	**ó-ŋmɛ̃̀ɛ̃́**	**ú-ꜜré**	**è-sádʒɛ̃̀**
	[ets]	-	-	**ù-bènè**	**ò-wɛ̀**	-	**ó-jɛ̀**	**ú-r̥ě**	-
	[une]	-	-	-	-	-	-	**ú-θèì**	-
	[aaa]	-	-	**ù-bèlè**	-	-	-	**ū-ré**	**ò-òdɛ̀**
	[oso]	-	-	**ù-βèrè**	**ò-wɛ̀**	**é-nɛrè**	-	-	**ò-zɛ̀**
d.	[atg]	-	-	-	**ò-wɛ̀**	-	-	**ú-lùé**	-
	[olm]	-	-	-	**ghɔ̂-mhìnhà**	-	**ghɔ́-hɔ̀**	**ú-thùnhámì**	**â-zɛ̀**
	[opa]	-	-	-	-	-	-	-	**à-dzɛ̀**
	[uha]	-	-	-	-	-	-	-	**ɛ̀-zɛ̀**
	[ehu]	-	-	-	-	-	-	-	**ɛ̀-rɛ̀**
	[uku]	-	-	-	-	-	-	-	**è-dɛ̀**

Lastly, there were a cognates which are proposed but rated with ‘low confidence’ (*n*=75), in (10). This rating was given to cognates which only exist for one Edoid branch, where cross-branch comparison is not possible. For these cognates, we can speculate or even infer the Proto-Edoid form, but further comparison would be needed for confirmation with another branch.

(10) Cognates with 'low confidence' (*n*=75)

		NAVEL_3	FOREST_3	BODY_3	HAWK_3	CHAIN	TOMORROW	SKIN_2	CORPSE_2
a.	[deg]	**ù-ɗúm**	**ɛ̀-dʒí**	-	-	-	-	-	-
	[enn]	**ù-ɗùmù**	**à-zhì**	-	-	-	-	-	-
	[epi]	**ù-ɗùmù**	**à-zì**	-	-	-	-	-	-
b.	[erh]	-	-	**ó-mà**	-	-	-	-	-
	[iso]	-	-	**ʊ̀-mà**	**ɔ̀-sʊ̀**	-	-	-	-
	[oke]	-	-	**ò-mà**	-	-	-	-	-
	[urh]	-	-	**ò-mà**	**ɔ̀-sò**	-	-	-	-
	[evh]	-	-	**ʊ̀-mà**	**ɔ́-sʊ̀**	-	-	-	-
c.	[bin]	-	-	-	-	**é-ɣȁ**	**á-xwɛ̀**	-	-
	[esa]	-	-	-	-	**ì-ɣȁ**	**á-xɔ̀**	-	-
	[ema]	-	-	-	-	**ì-ɣȁ**	**á-xɔ̀**	-	-
	[ets]	-	-	-	-	**à-ɣà**	**á-xwɛ̀**	-	-
	[une]	-	-	-	-	**ì-gà**	**á-xɔ̀**	-	-
	[aaa]	-	-	-	-	**ì-ɣà**	-	-	-
	[oso]	-	-	-	-	-	**à-kɔ̀**	-	-
d.	[atg]	-	-	-	-	-	-	-	-
	[olm]	-	-	-	-	-	-	**ghé-shì**	-
	[opa]	-	-	-	-	-	-	**wé-sì**	**ó-khùù**
	[uha]	-	-	-	-	-	-	**ì-sì**	**ò-gù**
	[ehu]	-	-	-	-	-	-	**ì-r̥ì**	**ō-gȕ**
	[uku]	-	-	-	-	-	-	**ì-r̥ì**	**ó-gȕ**

All three of these confidence grades – high, medium, and low – are included in the PEdT database (but denoted as such and can be included or excluded at the discretion of the user). At the same time, there were numerous words which were explicitly not included in the database, outside of this scale. The majority of these are known or suspected loanwords, for example the word for 'sea, ocean' in Edo and Emai is **òkṹ** which is nearly identical to the Yoruba [yor] form **òkũ̄**. Other excluded words are *wanderwörter* which are found widely in Southern Nigeria, e.g. the word for 'lion' which is **ò-dúmà** in Engenni, **ò-dùmã́** in Edo and Emai, and **ò-dùmhâ** in Etsako (all

Edoid languages), and also **òdúḿ** in Igbo [ibo] and **ìdùḿ** in Kalabari [ijn] (non-Edoid languages). A larger set of such excluded words is in (11), with comparison to Yoruba, various Igbo dialects, Ijoid languages (Mein Ijaw [ijc] and Kalabari), as well as one cultural word shared with Fongbe [fon] which is far from the Edoid area (spoken in the neighboring country of Benin).[6]

(11) Sample of excluded words suspected to be loanwords or *wanderwörter*

		'cutlass'	'gun'	'cow'	'forest spirit'	'net'	'knife'	'stone'
a.	[deg]	-	-	**nàm-búlò**	-	**ɔ́-ꜜgbɔ́**	-	**ʊ́-kʊ́tà**
	[enn]	-	-	-	-	**í-gbɔ́**	-	-
	[epi]	-	-	-	-	**ì-gbɔ́**	-	-
b.	[erh]	**ɔ̀-pìà**	-	-	-	-	-	-
	[iso]	**ɔ̀-pìà**	-	-	-	-	-	-
	[oke]	-	-	**námà**	-	-	-	-
	[urh]	**ɔ̀-pìà**	-	**ì-námà**	**à-zìzà**	-	-	**ù-kútā**
	[evh]	**ɔ̀-pìà**	-	-	-	-	-	-
c.	[bin]	**ɔ́-pìà**	**ó-sísí**	-	**ɛ̀-zìzà**	-	-	**ò-kútá**
	[esa]	**ɔ̀-pìà**	**ò-sísì**	-	-	-	-	-
	[ema]	**ɔ́-pìà**	**ò-íꜜsí**	-	**à-zìzà**	-	**à-gbòí**	-
	[ets]	**ɔ̀-pjà**	**ò-tsítsì**	-	-	-	-	-
	[une]	**ɔ̀-pjà**	-	-	-	-	-	-
	[aaa]	**ɔ̄-pjà**	-	-	-	-	-	-
	[oso]	**ɔ̀-pjà**	-	-	-	-	-	-
d.	[atg]	-	-	-	-	-	**à-gbòlí**	-
	[olm]	**ɔ́-pìà**	-	-	-	-	-	-
e.	Cf.							
	Yoruba	-	-	-	-	-	**àgbòrí** 'shaving knife'	**òkútā**
	Igbo	**òpíó** (Aro dialect)	**ósísí** 'tree, stick'	**námà** 'cattle'	-	**ìgbɔ́**	-	**òkwúté** (Onicha dialect)
	Ijoid	**àpìà** (Mein Ijaw)	-	**nàmbúlò** (Kalabari)	-	**íꜜgbɔ́** (Kalabari)	-	-
	Fongbe	-	-	-	**àzìzà** 'forest guardian'	-	-	-

[6] Additionally, we do not include numerals in this study ('two' through 'ten') despite the fact that they clearly show cognacy with regular sound correspondence (e.g. reflexes of 'four' in Degema **ìní**, Engenni **ínìì**, Urhobo **ɛ̀ɛ́nɛ̀**, Edo **ènɛ̃́**, Emai **èélè**, Ghotuo **èēnè**, Uhami **énè**, *etc*.). This is because numerals often show tonal quirks on the initial vowel which are not found on nouns, and these quirks are not fully understood at this point. We however do include larger numerals 'twenty' and 'thirty', which pattern as regular nouns.

4 Reconstructed nominal tone melodies for each Edoid branch

In this section, we present our reconstructions of the four individual branches, namely Delta Edoid (DE), South-West Edoid (SW), North-Central Edoid (NC), and North-West Edoid (NW). As stated, Elugbe (1989a) establishes this four-way split based on segmental reconstruction, but tone played no role in this proposal.

Of the PEdT database consisting of 436 cognates, the four branches range as to how many reflexes of these cognates are present. As shown in (12), this ranges from over 86% in NC but half that many in NW at just under 43%. Given that the NC branch as a whole is probably the best documented while NW is probably the worst, their high and low cognate percentages may be due simply to their varying documentation levels.

(12) Number of cognates found in each Edoid branch compared to total (*n*=436)

Branch	Cognates	Percentage of total
Delta (DE)	227	52.1%
South-West (SW)	297	68.1%
North-Central (NC)	377	86.5%
North-West (NW)	186	42.7%

As a preview, (13) summarizes the tonal correspondence sets within each branch: four in DE, five in SW, five in NC, and only two in NW. For each correspondence set, we provide our proposed tonal reconstruction. These are of the shape $*T_1$-$T_2(T_3)$, where T1 indicates the tone on the prefix and T_2 and T_3 (when present) indicate the tone of the stem. We refer to these reconstructions as 'nominal tone melodies'. As seen in this table, the number of cognates belonging to a tone melody is highly skewed: *L-L is by far the most common in DE, *L-L and *H-L both roughly equally dominant in SW and NC, and *H-L most common in NW.

(13) Summary of tonal correspondence sets and reconstructed nominal tone melodies

	Branch	Correspondence set	Reconstruction	*n*=
a.	Delta	DE-01	*L-L	125
		DE-02	*H-H	39
		DE-03	*H-L	19
		DE-04	*L-H	19
b.	South-West	SW-01	*L-L	91
		SW-02	*L-LH	38
		SW-03	*L-H	48
		SW-04	*H-L	78
		SW-05	*H-H	21
c.	North-Central	NC-01	*L-L	121
		NC-02	*H-L	118
		NC-03	*H-LH	53
		NC-04	*H-H	27
		NC-05	*L-H	27
d.	North-West	NW-01	*L-L	56
		NW-02	*H-L	103

Importantly, the nominal tone melodies constitute word-level patterns and are not intended to be reconstructions of individual tonemes. A reconstructed tone *H-L indicates that the prefix bore H tone and the stem bore a L tone which was distributed across the entire stem in some phonologically general way. The synchronic tone systems of Edoid languages tend to operate in terms of tone melodies in this manner, where the majority of nouns fall into a small number of patterns. For example, in Degema nearly 80% of all nouns fall into one of three word-level tone patterns (roughly, L, H, and LH – see Rolle & Kari 2022:259 for details).

4.1 Delta Edoid (DE)

The Delta Edoid (DE) sub-group contains three languages: Degema [`deg`], Epie-Atissa [`epi`], and Engenni [`enn`]. Degema is the best described, including a grammar (Kari 2004), a dictionary (Kari 2008), and a worked-out synchronic analysis of the tone system (Rolle & Kari 2022). We can establish 227 cognates showing reflexes in the DE branch, just over half the total number of Edoid cognates.

Within the available data, we can establish four primary tonal correspondence sets. The first and largest correspondence set is in (14), which we reconstruct as a Proto-DE nominal tone melody *L-L, and label it neutrally as 'DE-01' (*n*=125). This correspondence shows an all-low pattern in Engenni and Epie-Atissa, but a low plus final high pattern in Degema.

(14) Examples of DE-01, the *L-L melody (*n*=125)

	Cognate	deg	enn	epi	Proto-Edoid
a.	DOCTOR	**ɔ̀-ɓʊ́**	**ɔ̀-ɓʊ̀**	**ɔ̀-ɓʊ̀**	****O-ɓʊ**
	(PE_187)	L-H	L-L	L-L	
b.	GREY_HAIR	**ʊ̀-dáɲ**	**ɛ̀-dàà**	**ì-dàn**	****ghU-daN**
	(PE_281)	L-H	L-L	L-LL	
c.	ANIMAL_MEAT	**ɛ̀-nám**	**à-nàmʊ̀**	**à-nàmʊ̀**	****E-nhamhı**
	(PE_010)	L-H	L-LL	L-LL	
d.	ASH	**ə̀-məfú**	**ɛ̀-mʊ̀fʊ̀[mʊ̀]**	**ì-wòfù**	****A-mhuNəfu**
	(PE_023)	L-LH	L-LL[L]	L-LL	

These tonal transcriptions represent the surface pronunciation (e.g. as indicate in Elugbe 1989a, as well as Kari's 2008 dictionary). Crucially, Rolle & Kari (2022) show that this final H tone in Degema is actually not present in most contexts, and consequently they analyze this pattern as /L-L/ at a more abstract phonological level. Thus, the underlying forms in (14) would be akin to /**ɔ̀-ɓʊ̀**/ which surface as [**ɔ̀-ɓʊ́**]. This renders this tone melody as a whole as stable, not deviating from the proposed Proto-DE *L-L phonological representation in any daughter language.

Individual cognates may deviate from our expected correspondence, requiring us to posit either a separate correspondence set, or split the existing one into multiple subsets. In most cases here (and in what follows), we follow the latter strategy. For example, in (15) below cognates which show the expected all-low pattern are part of subset 'DE-01A' (by far the most numerous – *n*=69). However, there also exist a small minority of cognates for which the Engenni form has a high tone somewhere, which we label 'DE-01B' (*n*=3). Likewise, where Epie-Atissa has an unexpected H somewhere in the word are labeled 'DE-01C' and where Degema has a final L are

‘DE-01D’. We suspect these deviations are due to effects of a compound construction (either synchronically or historically).

(15) Subsets for DE-01 *L-L

Subset	*n*=	Cognate	deg	enn	epi
DE-01A	69	DEATH	**ù-wú** L-H	**ù-wù** L-L	**ù-wù** L-L
DE-01B	3	HAND_ARM	**ʊ̀-ɓɔ́** L-H	**ɔ́-ɓɔ̀** H-L	**ʊ̀-ɓɔ̀** L-L
DE-01C	7	BIRD	**ɛ̀-fɛ́n** L-H	**à-fɛ̀nì** L-LL	**ɛ̀-fɛ́ní** L-HH
DE-01D	7	BLOOD_2	**ì-zàlà** L-LL	**-**	**ì-zàrà** L-LL
DE-01U	39	BILE	**-**	**è-lò** L-L	**-**

The last subset ‘DE-01U’ are cognates which fit into multiple subsets due to the lack of relevant cognates in the other languages (here, ‘U’ stands for ‘unspecified’). Moreover, when a language has no data point – either because the word has been lost or simply hasn’t been documented in the literature – this is written with a dash ‘-’ in our tables.

The *L-L melody is as large as all the other classes combined. The first of these is a tone melody *H-H (DE-02 – *n*=39), exemplified in (16). In this set, Degema appears with a H-ꜜH pattern (with downstep), Engenni as H-L, and Epie-Atissa as L-HM. As above, some reflexes have an unexpected form: an initial H on the prefix in Epie-Atissa (DE-02B), or an all-low pattern in Engenni (DE-02C) or Epie-Atissa (DE-02D). Moreover, several cognates lack a form in Epie-Atissa, which prevents us from ascertaining its appropriate subset. We label these DE-02U (again, ‘U’ for ‘unspecified’). Lastly, those labeled DE-02X are miscellaneous patterns with other minor but unexpected deviations.

(16) Examples of DE-02, the *H-H melody (*n*=39)

Subset	*n*=	Cognate	deg	enn	epi
DE-02A	11	DOG_1	**á-ꜜɓʊ́wá** H-ꜜHH	**á-ɓʊ̀à** H-LL	**à-ɓʊ́ā** L-HM
DE-02B	5	RIVER_1	**é-ꜜdə́** H-ꜜH	**é-dà** H-L	**é-dē** H-M
DE-02C	3	WORM_1	-	**à-gìɔ̀** L-LL	**á-gān** H-M
DE-02D	2	SLAVE_2	**ó-ꜜkóní** H-ꜜHH	-	**ò-kònì** L-LL
DE-02U	12	BASKET_2	**ú-ꜜtúwó** H-ꜜHH	**ú-tùò** H-LL	-
DE-02X	6	MAN_1	-	**é-dèì** H-LL	**é-dībé** H-MH

The aforementioned Rolle & Kari (2022) study also establishes that Degema always inserts a downstep between two high tones at the end of a phonological phrase, and hence downstep is derived (…HH → …HꜜH). Given that downstep is derived, we may posit a simpler /H-H/ which surfaces as [HꜜH] in Degema when the word is phrase-final (e.g. RIVER_1 /**é-də́**/ → [**é-ꜜdə́**]). Two sound changes would then have taken place for the other DE languages. First, in Engenni the stem high becomes low (perhaps like Degema it was a derived downstepped high which then merged phonetically with low). Second, in most words in Epie-Atissa, the prefix high shifted rightward one syllable, a common diachronic effect from 'pitch delay' which we will see periodically in this study (§4.3). This results in the /L-HM/ forms transcribed in the sources (largely from Thomas & Williamson 1967, where we would interpret their M and Degema ꜜH as equivalent). At the same time, some words do not show the prefix L- and instead exhibit the same initial H as the other languages (e.g. DE-02B). At this point, no discernable conditioning environment has been found to explain the split here.

The final two DE correspondence sets are much smaller. The third set is reconstructed as a *H-L melody (DE-03 – *n*=19), largely based on data from Degema. In this set, Degema has a H-

HL (or H-L) form which corresponds to an initial H or L in Engenni but an initial L (if a prefix is present) in Epie-Atissa. However, for most of these cognates we only find a data point for Degema (DE-03U).

(17) Examples of DE-03, the *H-L melody (*n*=19)

Subset	*n*=	Cognate	deg	enn	epi
DE-03A	5	NOSE	**í-súβèŋ** H-HL	**ú-sùèì** H-LLL	**ù-súɓē** L-HM
DE-03B	2	TODAY	**í-nínə̀** H-HL	**ì-njâ** L-HL	**∅-jínēnē** ∅-HMM
DE-03U	12	COLD_N	**ó-nínì** H-HL	-	-

The fourth and final substantial set is reconstructed as *L-H, labeled DE-04 in (18). These have a L-HꜜH pattern in Degema (presumably from a H-tone breaking operation at some point), but a L-HL and L-HM pattern in Engenni and Epie-Atissa, respectively. Such patterns mirror the behavior of high tones in the *H-H series (DE-02) in (16) above: downstepped highs in Degema corresponds to low in Engenni and mid in Epie-Atissa.

(18) Examples of DE-04 *L-H (*n*=19)

Subset	*n*=	Cognate	deg	enn	epi
DE-04A	7	MORNING	**à-wíꜜjɛ́** L-HꜜH	**à-wíɛ̀** L-HL	**ɛ̀-wíɛ̄** L-HM
DE-04B	2	VILLAGE_1	**ò-wéꜜéj** L-HꜜH	**ò-gwè** L-L	-
DE-04U	5	NEEDLE_2	**ù-ɗíꜜɗíŋʷ** L-HꜜH	-	-
DE-04X	5	WING	**ʊ̀-βʊ́ꜜʊ́w** L-HꜜH	**ɛ́-vʊ̀ʊ̀** H-LL	**ʊ̀-vʊ̀** L-L

Our proposal for Proto-DE is summarized in (19), which provides the four correspondence sets with their reconstructed value, their canonical tonal reflex in the descendent languages, and the number and percentage from the total (*n*=227). One can see that the *L-L melody is by far the largest, followed by *H-H. Moreover, in Engenni and Epie-Atissa there are several mergers, which is indicative of the Delta branch as a whole. We will see this especially when we examine postulated changes from Proto-Edoid to Proto-DE (§5).[7]

(19) Four primary tonal correspondence sets in Delta Edoid

Proto-DE	Set	`deg`	`enn`	`epi`	Cognates	Total (*n*=227)
*L-L	DE-01	L-H	L-L	L-L	126	55.1%
*H-H	DE-02	H-ꜜH	H-L	L-HM/H-HM	39	17.2%
*H-L	DE-03	H-HL	H-L	L-HM	19	8.4%
*L-H	DE-04	L-HꜜH	L-HL	L-HM	19	8.4%
Other					25	11.0%

In (19), there are 25 cognates which are labeled 'other', which do not fit into these four sets but are not in sufficient numbers to warrant a separate correspondence set. For example, the cognate for TWENTY is [**ìjów**] in Degema, [**íjèì**] in Engenni, and [**ìjéè**] in Epie-Atissa, which is a L-H↔H-L↔L-HL correspondence not otherwise found. As it stands, these are interpreted simply as irregularities with the hope that future research can shed light on these forms (see §6.2 for extended discussion). Also included in 'other' are those cases where there is ambiguity where the cognates fits in because it is compatible with multiple correspondence sets, e.g. COUGH_N which only has a reflex in Epie-Atissa, [**ò-jílɔ̄**]. This form is compatible with *H-H, *H-L, or *L-H, and without a reflex form in Degema it is not possible to ascertain its categorization.

[7] We note that in DE, it may be the case that *H-H and *H-L can be collapsed into one proto-series, which would entail a split in Degema rather than a merger in the latter two DE languages. In Degema, trisyllable words show a near complementary distribution: [H-HL] words exist but [H-LL] are rare, and [H-ꜜHH] is allowed but [H-HꜜH] is not. This may speak to an earlier process where of *H-X (which could be either *H-H or *H-L, as there would be no contrast) become HꜜH or HL in Degema depending on syllable count.

4.2 South-West Edoid (SW)

Moving north, the next branch is South-West Edoid (SW). This contains five languages (sometimes culturally and linguistically subsumed within one another): Eruwa [erh], Isoko [iso], Okpe [oke][8], Urhobo [urh], and Uvwie [evh], of which Urhobo is the largest and best described. There were 297 cognates with reflexes in this branch, over two thirds of the total number of cognates (*n*=436).

The first correspondence class is reconstructed as *L-L (SW-01), in (20). For this melody, the majority of cognates for Eruwa show a L-H pattern, for Okpe a H-L pattern, and for the other three SW languages have L-L (SW-01A). In a significant minority, Eruwa has a L-L pattern and Okpe shows L-L as well (SW-01B and C). However, we must note that for the majority of cognates we have data only from Urhobo (SW-01U below), which is ambiguous on its own as to this patterning.

(20) Examples of SW-01 *L-L (*n*=91)

Subset	*n*=	Cognate	erh	iso	oke	urh	evh
SW-01A	19	TOOTH	**à-kɔ́** L-H	**à-kɔ̀** L-L	**á-wɔ̃̀** H-L	**à-kɔ̃̀** L-L	**à-kʊ̃̀** L-L
SW-01B	2	CHARM_DRUG_3	**ù-mù** L-L	**ù-mù** L-L	**ú-ʊ̃ʊ̃̀** H-L	**ù-xùʋù** L-LL	**ù-mù** L-L
SW-01C	10	FOOT_LEG_2	**à-ʋɔ̀** L-L	**ʊ̀-wɔ̀** L-L	**ò-wɔ̀** L-L	**ò-wɔ̀** L-L	**à-ʋɔ̀** L-L
SW-01D	8	FEAR_2	**-**	**ʊ̀-z̈ɔ̀** L-L	**-**	**ò-ʃɔ̀** L-L	**ò-ɾ̥ɔ̀** L-L
SW-01U	48	BALD_N	**-**	**-**	**-**	**à-kpã̀** L-L	**-**
SW-01X	4	BODY_3	**ó-mà** H-L	**ʊ̀-mà** L-L	**ò-mà** L-L	**ò-mà** L-L	**ʊ̀-mà** L-L

[8] Note that Okpe with ISO code [oke] is different from the identically named Edoid language Okpe with ISO code [okx]. Okpe-[oke] belongs to the South-West branch and is surveyed here, while Okpe-[okx] belongs to the North-West branch and is not surveyed in this study (see Lewis 2013 for some data on Okpe-[okx]).

Despite the presence of H tones in Eruwa and Okpe, we reconstruct this as *L-L in Proto-SW for three reasons. First, this is the pattern actually seen in a minority of cognates (SW-01C above). Second, there is variation across the Eruwa and Okpe sources between a [L-L] form and one with a high tone, for example Eruwa BAG_1 is [**ɛ̀-kpá**] ~ [**ɛ̀-kpà**], EYE is [**à-rʊ́**] ~ [**à-rò**], LOUSE_1 is [**ù-zú**] ~ [**ù-zù**], and AXE_1 is [**è-zúé**] ~ [**è-zùè**], depending on the source. Third, we see several places in Edoid where all-low patterns add an epenthetic high tone as a synchronic rule in the language (seen already in Degema in §4.1, and to be seen in Oloma in §4.4). We therefore posit a rule which added a final H to all-low patterns in Eruwa and an initial H in Okpe in the same context. It is unclear if this is an active alternation in these languages, or what conditions its non-application in a minority of cases (SW-01C in (20)).[9]

There are two other sizable correspondence sets which involve an initial L tone on the prefix: one reconstructed as *L-LH (SW-02) and one as *L-H (SW-03). These are exemplified in (21) and (22), respectively. For the *L-LH melody, Eruwa and Okpe have a final high tone while Isoko, Urhobo, and Uvwie have all low tones (with some minor subsets, as indicated). This distribution mimics that of *L-L (SW-01) in in (20) above where it was also Eruwa and Okpe which exhibit a high tone somewhere. The motivation for analyzing SW-02 as *L-LH (while SW-01 as *L-L) is that this SW-02 class corresponds to cognates with a high tone in NC and NW Edoid branches. In contrast, SW-01 primarily corresponds to all low-toned in those same branches.

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[9] One potential explanation may involve the fortis/lenis distinction from Proto-Edoid (since broken down in SW). For example, words belonging to SW-01A which condition the H-addition rule include Proto-Edoid TOOTH ****dhI-kʊN**, GREY_HAIR ****ghU-daN**, SUN ****U-vuNə**, WATER ****A-mɪN**, and WAR_FIGHT_2 ****dhI-khʊmhɪ**. The last of these cognates is the only one with a reconstructed lenis consonant (the voiceless velar stop ****kh**). In contrast, words of the SW-01C subset which shows all-low patterns across SW languages are more prone to have an initial lenis consonant, e.g. EYE ****dhI-dhʊ**. However, reconstructed fortis consonants also exist here (e.g. LIFE_1 ****A-gbɔN**).

(21) Examples of SW-02 *L-LH (*n*=38)

Subset	*n*=	Cognate	erh	iso	oke	urh	evh
SW-02A	27	MOUTH	**è-nú** L-H	**ù-nù** L-L	**ù-nú** L-H	**ù-nù** L-L	**è-nù** L-L
SW-02B	5	TONGUE	**ò-rɔ̀ʋó** L-LH	**ɛ̀-rɔ̀ɔ̀** L-LL	**ò-rɛ́mɔ̀** L-HL	**ɛ̀-ɻèβɛ̀** L-LL	-
SW-02C	2	DOCTOR	**ɔ̀-bɔ̀** L-L	**ɔ̀-bɔ̀** L-L	**ɔ̀-bó** L-H	**ɔ̀-bò** L-L	**ɔ̀-bɔ̀** L-L
SW-02X	4	NAVEL_2	**ú-tùlɛ̀** H-LL	**ù-tùlɛ̀** L-LL	**è-r̥úr̥ɛ̀** L-HL	-	**è-r̥úr̥ù** L-HL

From these data, we must posit a diachronic change which merged the *L-LH and *L-L melodies in Isoko, Urhobo, and Uvwie, where the final H was lost. In Eruwa and Okpe, this changed to L-H and therefore merged with the *L-H melody (immediately below).

Another correspondence set SW-03 is reconstructed as a *L-H melody, which remains an unchanged L-H pattern across all five SW languages (with minor variations). Note that we also include in the *L-H melody cognates which have L-HL across the SW languages (SW-03B below), in order to have a large enough number for comparison to other Edoid branches later (§5).

(22) Examples of SW-03 *L-H (*n*=48)

Subset	*n*=	Cognate	erh	iso	oke	urh	evh
SW-03A	28	GOAT_1	**ɛ̀-ʋí** L-H	**ɛ̀-ʋí** L-H	**ɛ̀-ʋé** L-H	**ɛ̀-ʋé** L-H	**ɛ̀-ʋí** L-H
SW-03B	6	ELDER_4	**ɔ̀-kpákɔ̀** L-HL	**ɔ̀-kpákɔ̀** L-HL	**ò-kpákò** L-HL	**ò-kpákò** L-HL	**ɔ̀-kpákɔ̀** L-HL
SW-03C	3	HUNTER	-	**ó-zúɛ́** H-HH	**ò-r̥úɛ́** L-HH	**ò-r̥úɛ́** L-HH	**ò-r̥úɛ̀** L-HL
SW-03X	11	KNEE	-	**ù-gbó** L-H	**ù-gbó** L-H	**ù-gbò** L-L	-

Next, let us examine our final two major correspondence sets, both of which are constructed with an initial H on the prefix: a *H-L melody (SW-04) and a *H-H melody (SW-05). Compared to the L-initial reconstructed forms above, the H-initial forms show a higher degree of variation in the descendent forms, making it much more difficult to ascertain what should be treated as a subset from what should be treated as a new correspondence set. First consider the *H-L melody (23), which has a H-L pattern across all SW languages (SW-04A). However, a sizeable minority of cognates in Urhobo (SW-04B) instead show a H-ꜜH correspondent containing a downstepped high (alternatively transcribed as H-M), which is roughly half the size of SW-04A. An example is the reflex of GROUND, which in Urhobo is /**óꜜtɔ́**/. At this point, we classify these as a single melody *H-L (as opposed to two classes *H-L and *H-ꜜH which merged in all SW but Urhobo). We do so because the /H-L/ words versus the /H-ꜜH/ words in Urhobo do not cleanly correspond to distinct sets in other Edoid branches. Moreover, it is previously known in SW that /H-L/ patterns can be realized a [H-ꜜH] when there is no contrast, a process which is optional but synchronically active in Isoko (Elugbe 1977). At this point, we simply note this discrepancy in Urhobo and await further study of a possible conditioning factor (or evidence for borrowing or dialect mixing, perhaps).[10]

[10] It should perhaps also be noted that even in Urhobo, the /H-ꜜH/ words are realized as [H-L] when they are in non-phrase final position (Welmers 1969).

(23) Examples of SW-04 *H-L (n=78)

Subset	n=	Cognate	erh	iso	oke	urh	evh
SW-04A	33	SMOKE_1	-	**í-ʋìrrì** H-LL	**í-ìrĩ̀** H-LL	**í-ʋìlì** H-LL	**í-ʋìrì** H-LL
SW-04B	17	GROUND	**ʊ́-rɔ̀rì** H-LL	**ʊ́-tɔ̀** H-L	**ó-tɔ̀rì** H-LL	**ó-ꜜtɔ́** H-ꜜH	**ó-r̥ɔ̀** H-L
SW-04C	4	BONE_1	**ù-βɛ̀** L-L	**ú-ŋwè** H-L	**ú-ʋɛ̃̀** H-L	**í-ꜜwé** H-ꜜH	**ú-wɛ̃̀** H-L
SW-04D	5	YESTERDAY	**ó-dɛ̀** H-L	**ò-dɛ̀** L-L	**ó-dɛ̀** H-L	**ó-ꜜdɛ́** H-ꜜH	-
SW-04E	6	ROPE_2	**ú-fì** H-L	**ú-fì** H-L	**ù-fí** L-H	**ú-ꜜfí** H-ꜜH	**ú-fì** H-L
SW-04U	7	URINE_2	**á-kà** H-L	-	**é-kà** H-L	-	**á-kà** H-L
SW-04X	6	SHELL	-	**ú-ɣó** H-H	**í-ɣò** H-L	**í-ꜜɣó** H-ꜜH	-

The other, much smaller H-initial set is reconstructed as *H-H, in (24). In this set, Isoko and Urhobo retain the /H-H/ pattern, while the other three SW languages have condensed the high span to a single position, either to the prefix (in Eruwa) or the stem (Okpe and Uvwie). However, like we saw with *H-L (23) there are frequent deviations from the expected reflexes (e.g. lowering in Isoko in SW-05B, *etc.*)

(24) Examples of SW-05 *H-H (n=21)

Subset	n=	Cognate	erh	iso	oke	urh	evh
SW-05A	6	TREE	**ʊ́-rã̀rɛ̃̀** H-LL	**ú-ré** H-H	**ò-r̥ã́** L-H	**ú-r̥é** H-H	**ò-rá** L-H
SW-05B	3	HEART	**[ú-ví]-ú-dù** [H-H]-H-L	**ú-dù** H-L	-	**ú-dú** H-H	-
SW-05U	5	TRAP	-	-	-	**ú-wɛ̃́** H-H	-
SW-05X	7	EGG	**í-kè** H-L	**é-ké** H-H	**ì-jɛ̃́** L-H	**ú-kɛ̃́** H-H	**ú-kɛ̃́** H-H

We summarize the five SW correspondence sets in (25).

(25) Five major correspondence sets in South-West Edoid

Proto-SW	Set	erh	iso	oke	urh	evh	Cognates	Total (*n*=297)
*L-L	SW-01	L-H/L-L	L-L	H-L/L-L	L-L	L-L	91	30.6%
*L-LH	SW-02	L-H	L-L	L-H	L-L	L-L	38	12.8%
*L-H	SW-03	L-H	L-H	L-H	L-H	L-H	48	16.2%
*H-L	SW-04	H-L	H-L	H-L	H-L/H-ꜜH	H-L	78	26.3%
*H-H	SW-05	H-L	H-H	L-H	H-H	L-H	21	7.0%
Other							21	7.0%

4.3 North-Central Edoid (NC)

Moving further north, the next branch is North-Central Edoid (NC) which includes the languages Edo [bin], Esan [ish], the Emai lect of the Emai-Ora-Iuleha cluster [ema], Etsako [ets], Uneme [une], Ghotuo [aaa], and Ososo [oso]. The largest portion of cognates appear in this branch, 377 of the 436 total. Notably, the NC branch contains the Edo language (also known as Bini), the language of the Benin Empire which dominated this region of Nigeria politically, militarily, and culturally from the 15th to 19th centuries (Egharevba 1968 [1934]; Ryder 1969).

Two correspondence sets together account for over 50% of NC cognates: one reconstructed as *L-L (set NC-01, *n*=121) and one reconstructed as *H-L (set NC-02, *n*=119). The proto-NC *L-L melody is exemplified in (26), where all seven NC languages retain the all-L pattern (NC-01A). For each NC language, however, a high tone may sporadically appear, typically on the prefix (NC-01B to NC-01H). The most notable of these is Edo (subset NC-01B), which have the largest number of words with an initial H tone (*n*=14).

(26) NC-01 *L-L (*n*=121)

Subset	*n*=	Cognate	bin	ish	ema	ets	une	aaa	oso
NC-01A	59	MARKET	**ɛ̀-kì** L-L	**è-kì** L-L	**è-kì̃** L-L	**à-kì** L-L	**à-kì** L-L	**ɣò-kì** L-L	**ò-xì** L-L
NC-01B	14	CHAIN	**é-ɣà̃** H-L	**ì-ɣà̃** L-L	**ì-ɣà̃** L-L	**à-ɣà** L-L	**ì-gà** L-L	**ì-ɣà** L-L	-
NC-01C	2	FRIEND_1	**ɔ̀-sè** L-L	**ɔ́-siɔ̀** H-L	-	-	-	**ɔ̀-sè** L-L	**ɔ̀-tʃì[àrì]** L-L[LL]
NC-01D	4	AGE_GRADE_1	**ò-tù** L-L	**ò-tù** L-L	**ò-tú** L-H	**ò-tù** L-L	**ò-tù** L-L	-	-
NC-01E	1	FOREST_2	-	-	**ò-gò** L-L	**ó-gwà** H-L	-	-	**è-gwà** L-L
NC-01F	1	PROFIT	**è-rè** L-L	-	**è-è** L-L	**è-lèlè** L-LL	**ê-lé** F-H	**è-lèlì** L-LL	-
NC-01G	2	JUJU_MAGIC	**ɛ̀-bɔ̀** L-L	-	-	-	-	**ɛ̄-wɔ̀** M-L	**[ìz]ò-bò** [L]L-L
NC-01H	4	SUN	**ò-vɛ̃̀** L-L	**ò-vwɛ̃̀** L-L	**ò-vɔ̃̀** L-L	**ò-vɔ̀ɔ̀** L-LL	**ò-vɔ̀r̃ì** L-LL	**ò-vɔ̀** L-L	**ó-vɔ̀** H-L
NC-01U	20	BOUNDARY_3	-	-	**ù-hì** L-L	-	-	-	-
NC-01X	14	TIME	**ɛ̀-ɣɛ̀** L-L	**ɛ́-ɣèlè** H-LL	**ɛ́-ɣɛ̀** H-L	**ɛ̀-ɣɛ̀ɣɛ̀** L-LL	**ɛ̀-lɛ̀ɣɛ̀lì** L-LLL	-	-

The second largest set is the *H-L melody, in (27). While the reflex of this is H-L in most NC languages, in Ghotuo [aaa] it is M-L (which contrasts with high tone), and in Edo it appears to have shifted one syllable rightwards (i.e. *H-L > L-H). This reconstruction entails a 'Great Edo Tone Shift' whereby a formerly initial high tone on the prefix moved onto the stem.[11]

[11] The 'Great Edo Tone Shift' is an allusion to similarly named diachronic tone changes in the literature, such as the 'Great Igbo Tone Shift' (Hyman 1974) and the 'Great Ngamo Tone Shift' (Schuh 2017:135).

(27) Reflexes of NC-02 *H-L (*n*=119)

Subset	*n*=	Cognate	bin	ish	ema	ets	une	aaa	oso
NC-02A	47	DOG_1	**à-wá** L-H	**á-wà** H-L	**á-wà** H-L	**á-ɣwà** H-L	**á-wà** H-L	**ɣā-wà** M-L	**á-wà** H-L
		HEAD	**ù-húʋũ̀** L-HL	**ú-hɔ̃̀βɔ̃̀** H-LL	**ú-hũ̀mì** H-LL	**ú-sò** H-L	**ú-ʃòmhì** H-LL	**ū-sɔ̀** M-L	**ú-kwɛ̀** H-L
NC-02B	16	CHICKEN	**ɔ̀-xɔ́xɔ̀** L-HL	**ɔ́-xɔ́xɔ̀** H-HL	**ɔ́-ɔ́xɔ̀** H-HL	**ɔ́-xɔ́xɔ̀** H-HL	**ɔ́-xɔ́xɔ̀** H-HL	**ɔ̄-hɔ̀** M-L	**ɔ́-xɔ̀xɔ̀** H-LL
NC-02C	11	CLOTH_1	**ù-kpɔ̃̀** L-L	**ú-kpɔ̃̀** H-L	**ú-kpũ̀** H-L	**ú-kpò** H-L	**ú-kpò** H-L	**ū-kpò** M-L	-
NC-02D	2	NECK_ THROAT_2	**ɛ̀-hóò** L-HL	**à-xɔ̀lè** L-LL	-	-	-	**ɛ́-hàlì** H-LL	-
NC-02E	2	BLOOD_1	**è-r̥ɛ́ɛ̃̀** L-HL	**á-rã̀lɛ̃̀** H-LL	**ɛ̀-rèè** L-LL	**á-r̥àì** H-LL	**í-θàì** H-LL	-	-
…									
NC-02U	9	GIFT	**ɔ̀-hɛ́** L-H	-	-	-	-	-	-
NC-02X	23	SALT	**ù-ʋɛ́ɛ̃̀** L-HL	**ú-βɛ̃̀lɛ̃̀** H-LL	**ú-mɛ̀ɛ̀** H-LL	**ú-mhèè** H-LL	**ú-mhèlì** H-LL	-	**ù-bú** L-H

The expected outcome for *H-L involving multisyllabic stems is as with HEAD above: L-HL in Edo and H-LL in those NC languages which retain the bisyllabic stem structure. At the same time, a number of cognates pattern as in NC-02B CHICKEN, where the reflex is H-HL. While it is reasonable that this should be a distinct correspondence set – perhaps a reconstructed *H-HL value – we nonetheless collapse it with NC-02A, as we shall see that they pattern alike in our cross-Edoid comparison. This entails that in multisyllabic stems, a prefixal H spreads onto a low-stem in some (but not all) cognates. Moreover, a number of cognates have an unexpected all-low pattern appears in one language. In the table, we divide this by whether the all-low pattern occurs in Edo (NC-02C), Esan (NC-02D), Emai (NC-02E), *etc*. (those of NC-02F through NC-02I, not shown, have five or fewer members).

At the same time, we can also establish a tonal correspondence set where the initial H in Edo seems not to have shifted, where all NC languages begin with a prefixal H-. This is correspondence set NC-03 which we reconstruct as a *H-LH melody, in (28). This is split into two equally numerous subsets: one with an initial H followed by a L in the stem somewhere with no additional H tone (NC-03A), versus one with a H-L plus an additional H in one or more NC languages (NC-03B). The first subset NC-03A mostly consists of two-syllable nouns (i.e. monosyllabic stems), while the second NC-03B entirely consists of three-syllable ones.

(28) Reflexes of NC-03 *H-LH (*n*=54)

Subset	*n*=	Cognate	bin	ish	ema	ets	une	aaa	oso
NC-03A	18	STONE_1	**ú-dò** H-L	**ú-dò** H-L	**ú-dò** H-L	**ú-dò** H-L	**ú-dò** H-L	**ū-dò** M-L	-
		TONGUE	**á-ɽɛ̃́ɱɛ̃̀** H-HL	**[úkp]é-làβɛ̃̀** [H]H-LL	**ó-ɛ̃̀mì̃** H-LL	**ó-lɛ̀mhì** H-LL	**ɛ́-r̃ɛ̀mhì** H-LL	**ɣī-ɹɛ̀** M-L	**í-rɛ̀r̥ɛ̀** H-LL
NC-03B	19	COCOYAM	**í-jòxó** H-LH	**í-jòxô** H-LF	**í-xuòxuó** H-LH	**í-jàxô** H-LF	**í-jòxô** H-LF	-	-
NC-03C	6	ANT_ TERMITE_1	**é-dɔ̃̀** H-L	**é-dɔ̃̀** H-L	**é-ꜜdó** H-ꜜH	**ó-dò** H-L	**í-dòdô** H-LF	-	-
NC-03U	2	BAG_2	**ɛ́-kpò** H-L	-	-	-	-	-	-
NC-03X	9	ROAD_1	**ú-ɣè** H-L	**ú-ɣè** H-L	-	**ù-gjè** L-L	-	-	-

The H-LH reflex of NC-03B suggest that the Great Edo Tone Shift was blocked if the prefixal H would've become next to a stem H. We may therefore assume that there was a former H tone at the right edge of words in the *H-LH melody, which became unlinked and ultimately deleted under certain conditions in NC-03A, such as when there were not enough syllables to host the number of tones. This final H can account for the non-appearance of the Great Edo Tone Shift in NC-03A, even though the conditions appear to have been met. This is further supported by a small number of words (largely from Emai) which show a HꜜH pattern in this set too (in NC-03C), which one could take to be another reflex of the *H-LH melody.

The last two NC correspondence sets are the smallest: one reconstructed as *H-H (NC-04) and one reconstructed as *L-H (NC-05). For the *H-H melody, we take the core reflex to be a H-

H pattern retained only in the Edo language, which in all other NC languages becomes H-L (lowering of the stem), shown in (29) (specifically NC-04A). However, it is also common for the H-H in Edo to correspond to a H-LH pattern (NC-04B) or H-ꜜH (NC-04C) in one or more of the other NC languages. Less common are instances which correspond to an initial L- in one of these other NC languages (NC-04D).

(29) Reflexes of NC-04 *H-H (*n*=27)

Subset	*n*=	Cognate	bin	ish	ema	ets	une	aaa	oso
NC-04A	6	WING	**í-fwɛ̃́** H-H	**í-hwɛ̃̀** H-L	**í-fwà** H-L	**í-fwà** H-L	**[áb]ì-fò** [H]L-L	-	**í-fwà** H-L
NC-04B	3	NAIL_1	**é-hjɛ̃́** H-H	**é-hjɛ̃̀lɛ̃̀** H-LL	**é-hìɛ̃́** H-LH	**é-fjɛ̀** H-L	**é-fjɛ̀r̃ì** H-LL	**ɛ̄-ɛ̄fwɛ̀** M-ML	**é-fjà** H-L
NC-04C	7	SHELL	**í-ɣó** H-H	**í-ꜜɣô** H-ꜜF	**é-ꜜɣó** H-ꜜH	**[íkp]í-ɣó** [H]H-H	**[íkp]í-ɣò** [H]H-L	**ē-ɣō** M-M	-
NC-04D	6	COURT_CASE	**έ-zɔ́** H-H	-	**ɛ̀-zɔ̃́** L-H	**ɛ̀-dzɔ̂** L-F	**ɛ̀-zɔ̂** L-F	-	-
NC-04U	2	SWORD_2	**á-dá** H-H	-	-	-	-	-	-
NC-04X	3	CRAB	**ò-zì** L-L	**ó-zì** H-L	**ó-ꜜzí** H-ꜜH	**ó-dzî** H-F	**ó-dʒí** H-H	-	-

Such patterns suggest that while H-H is faithfully retained in Edo, there was a systematic change throughout NC (though whose output was conditioned by indeterminate factors at this point). Such a change accounts for the fact that H-H is systematically banned as a lexical tone pattern in all but Edo in this Edoid branch. There are no examples of /H-H/ words in Esan, Emai, or Ghotuo (and only two M-M), only one /H-H/ in Etsako and Uneme, and only two in Ososo.

Lastly, the reconstructed *L-H melody is exemplified in (30). In these data, all NC languages begin with a L-H sequence, with or without an additional low tone word-finally (NC-05A). A minority of cognates show a downstepped high in Emai (NC-05B), an all-low pattern in one or more NC languages (NC-05C), or some other unexpected value such as an initial H (NC-05X).

(30) Reflexes of NC-05 *L-H (*n*=27)

Subset	*n*=	Cognate	bin	ish	ema	ets	une	aaa	oso
NC-05A	12	HOE_1	**ɛ̀-gwɛ́** L-H	**ɛ̀-gwɛ̂** L-F	**ɛ̀-gúɛ́** L-HH	**ɛ̀-gúɛ̀** L-HL	**ɛ̀-gúɛ̀** L-HL	**ɛ̀-gbā** L-M	**ɛ̀-gúɛ́** L-HH
NC-05B	4	BLOOD_2	**è-ságjɛ̏** L-HL	-	**è-sádʒɛ̏** L-HL	-	-	**ò-òdɛ̀** L-LL	**ò-zɛ̀** L-L
NC-05C	2	CROCODILE_4	**ɛ̀-ɣúɣù** L-HL	-	**ɛ̀-ɣúꜜɣú** L-HꜜH	-	-	-	-
NC-05U	6	HOUSE_5	-	-	-	**ò-dî** L-F	-	-	-
NC-05X	3	MAT_3	**è-wá** L-H	-	**é-wáà** H-LL	**ì-júíá** L-HHH	**ì-wɔ̂** L-F	-	-

We summarize the five reconstructed tone classes for NC in (31).

(31) Five major correspondence sets in North-Central Edoid (total *n*=377)

P-NC	Set	bin	ish	ema	ets	une	aaa	oso	*n*=	Percentage
*L-L	NC-01	L-L/H-L	L-L	L-L	L-L	L-L	L-L	L-L	121	32.1%
*H-L	NC-02	L-H(L)	H-L	H-L	H-L	H-L	M-L	H-L	119	31.6%
*H-LH	NC-03	H-L(H)	H-L(H)	H-L(H)	H-L(H)	H-L(H)	M-L	H-L(H)	53	14.1%
*H-H	NC-04	H-H	H-L(H)	H-L(H)	H-L(H)	H-L(H)	M-L	H-L(H)	27	7.2%
*L-H	NC-05	L-H(L)	L-H(L)	L-H(L)	L-H(L)	L-H(L)	L-M(L)	L-H(L)	27	7.2%
Other									30	8.0%

4.4 North-West Edoid (NW)

Finally, the North-West branch of Edoid is in the far north of the Edoid cultural complex, an area with rolling hills closer to the posited homeland for the Proto-Edoid people (Elugbe 1979; Lewis 2013), and even Proto-Benue-Congo itself (Williamson 1989). Of the four Edoid branches, we have the sparsest data on the North-West branch, with only 186 cognates showing reflexes (out of the 451 total). We present data from 6 languages: North Ibie [atg], Oloma [olm], Okpamheri [opa], Uhami [uha], Ehueun [ehu], and Ukue [uku].

In general, NW Edoid is remarkable for its lack of tone classes. Most nouns can be placed in one of two correspondence sets. The first is reconstructed as the now familiar *L-L melody (NW-

01), which in all but one NW language is retained as L-L, in (32). The only location of significant variation is in Oloma, which for most cognates are transcribed with an additional high at the left edge resulting in a falling tone on the prefix (NW-01A). The addition of a high tone to an all-low pattern is parallel to that seen already in Degema (§4.1), but where a H was added to the right edge. Even in Oloma, however, there are still tokens transcribed with L-L (NW-01B), or where an Oloma cognate is absent and sorting into one of the two subsets cannot be determined (NW-01U). A small number of cognates have an initial H in one or more NW language as well, where all other NW languages retain the L-L (NW-01X).

(32) Reflexes of NW-01 *L-L (*n*=56)

Subset	*n*=	Cognate	atg	olm	opa	uha	ehu	uku
NW-01A	20	TOOTH	**à-kò** L-L	**lɛ̂-kɔ̀** F-L	**rà-kɔ̀** L-L	**à-kɔ̀** L-L	**à-kȕ** L-L	**à-kȕ** L-L
NW-01B	6	CHIN	**è-kpàmì** L-LL	**à-gbà** L-L	**à-gbà** L-L	**à-gbàmì** L-LL	**à-wȁ** L-L	**à-gbȁ** L-L
NW-01U	20	SOIL_1	**è-kɛ̀** L-L	**-**	**ɛ̀-kɛ̀** L-L	**è-kè** L-L	**è-kɛ̏** L-L	**è-kɛ̏** L-L
NW-01X	10	FAECES_1	**-**	**ì-sɔ̀** L-L	**ì-sò** L-L	**í-sɔ̀** H-L	**ì-sɔ̏** L-L	**ì-ɹ̥ɔ̏** L-L

The second correspondence NW-02 is reconstructed as a *H-L melody (33). The reflex of this class is H-L in all NW languages except Ehueun where it is mostly M-L (NW-02A), but even here occasionally H-L (NW-02B). There exist a small number of subsets involving an all-low pattern in one or more NW language (NW-03C), a rising word-level contour (e.g. L-H or M-H – NW-03D), or a H-LH pattern (NW-03E).

(33) Reflexes of NW-02 *H-L (*n*=103)

Subset	*n*=	Cognate	atg	olm	opa	uha	ehu	uku
NW-02A	25	HEAD	**ú-xòmhì** H-LL	**ghí-kà** H-L	**ú-khèmhì** H-LL	**ú-kèmì** H-LL	**ū-hȕ** M-L	**ú-kɔ̀mì** H-LL
NW-02B	4	MOUTH	-	**ú-nù** H-L	**ú-nù** H-L	**ú-nù** H-L	**ú-nù** H-L	**ú-nù** H-L
NW-02C	15	CORPSE_2	-	-	**ó-khùù** H-LL	**ò-gù** L-L	**ō-gȕ** M-L	**ó-gȕ** H-L
NW-02D	8	NIGHT_1	-	**ghá-úzɛ̀** H-HL	**á-zʊ̀** H-L	**ɛ̀-dùzí** L-LH	**ū-dʒɨ́** M-H	**á-ɽ̥ȕ** H-L
NW-02E	3	SNAIL	**ú-lùé** H-LH	**ú-thùnhámì** H-LHL	-	-	-	-
NW-02U	41	MOTHER_1	-	-	-	-	**ō-nȕ** M-L	-
NW-02X	7	ANIMAL_ MEAT	**ɛ́-làmì** H-LL	**â-nhà** F-L	**ā-r̃àmhì** M-LL	**ɛ́-námì** H-HL	**ɛ̄-nàȕ** M-LL	**ɛ́-nàmì** H-LL

We summarize the two reconstructed tone classes for NW in (34). Under 'Other' are those which do not fit into these correspondence sets in any clear way. The largest of these is a tentative set of L-H nouns which appear in one or more NW language (e.g. HOUSE_5 **òdé** in Uhami and Ukue, and **ōdé** in Ehueun), though these do not occur in enough numbers so as to do comparison to other branches.

(34) Two major correspondence sets in North-West Edoid (total *n*=186)

P-NC	Set	atg	olm	opa	uha	ehu	uku	*n*=	Percentage
*L-L	NW-01	L-L	F-L/L-L	L-L	L-L	L-L	L-L	56	30.1%
*H-L	NW-02	H-L	H-L	H-L	H-L	M-L	H-L	103	55.4%
Other								27	14.5%

4.5 Summary of proposed tone changes

Compiling the proposals from all four Edoid branches, we are left with the following correspondence sets, proposed proto-melodies, and reflexes in the descendent languages, in (35). In this table, tone changes which are proposed to have taken place from the proto-stage are in red.

(35) All major correspondence sets across Edoid and their common reflexes (changes in red)

	Set	Proto							
	P-DE set		deg	enn	epi				
a.	DE-01	*L-L	> L-**H**	L-L	L-L				
b.	DE-02	*H-H	> H-**ꜜ**H	H-**L**	**L**-H**M**/H-H**M**				
c.	DE-03	*H-L	> H-**H**L	H-L	**L**-H**M**				
d.	DE-04	*L-H	> L-H**ꜜH**	L-H**L**	L-H**M**				
	P-SW set		erh	iso	oke	urh	evh		
e.	SW-01	*L-L	> L-**H**/L-L	L-L	**H**-L/L-L	L-L	L-L		
f.	SW-02	*L-LH	> L-**H**	L-**L**	L-**H**	L-**L**	L-**L**		
g.	SW-03	*L-H	> L-H	L-H	L-H	L-H	L-H		
h.	SW-04	*H-L	> H-L	H-L	H-L	H-L/H-**ꜜH**	H-L		
i.	SW-05	*H-H	> H-**L**	H-H	**L**-H	H-H	**L**-H		
	P-NC set		bin	ish	ema	ets	une	aaa	oso
j.	NC-01	*L-L	> L-L/**H**-L	L-L	L-L	L-L	L-L	L-L	L-L
k.	NC-02	*H-L	> **L**-H(L)	H-L	H-L	H-L	H-L	**M**-L	H-L
l.	NC-03	*H-LH	> H-L(H)	H-L(H)	H-L(H)	H-L(H)	H-L(H)	**M**-L	H-L(H)
m.	NC-04	*H-H	> H-H	H-**L**(H)	H-**L**(H)	H-**L**(H)	H-**L**(H)	**M-L**	H-**L**(H)
n.	NC-05	*L-H	> L-H(L)	L-H(L)	L-H(L)	L-H(L)	L-H(L)	L-**M**(L)	L-H(L)
	P-NW set		atg	olm	opa	uha	ehu	uku	
o.	NW-01	*L-L	> L-L	**F**-L/L-L	L-L	L-L	L-L	L-L	
p.	NW-02	*H-L	> H-L	H-L	H-L	H-L	**M**-L	H-L	

We can group tone changes into various types. First, independently in at least three Edoid branches there has been a rule which adds a high tone to an all-low noun, what we can call 'H-Obligatoriness' (using the term from Hyman 2006). In Delta Edoid, *L-L becomes L-H in Degema [deg] in words in isolation, which is a synchronically active rule in the language (Rolle & Kari 2022). In South-West Edoid, *L-L becomes L-H in Eruwa [erh] and H-L in Okpe [oke] in a great many cognates (though others remain L-L). The status of this as a purely diachronic change or as a synchronically active rule is not known presently. And in Oloma [olm] in North-West Edoid, most *L-L words have become F-L.

Another type of recurring change is lowering a H tone to a downstepped ꜜH, a mid M, or a low L, referred to as 'H-Lowering'. In some instances, all H's lower to M regardless of position, such as in the NC language Ghotuo [aaa] or the NW language Ehueun [ehu], likely due to the development of a higher contrastive pitch. In other cases, H-lowering took place only after a H tone, whether to downstep (Degema [deg]), mid (Epie-Atissa [epi]), or low (Engenni [enn]; Eruwa [erh]; almost all of NC Edoid). The reverse process of 'L-Raising' is much less common, and is found only in Urhobo [urh] where *H-L becomes either H-L or H-ꜜH. Even here, it can be debated whether this is an innovation or a retention.

Another common change is 'Rightward H-Shift', best exemplified by what we called the Great Edo Tone Shift whereby Proto-NC *H-L becomes L-H(L) in Edo [bin]. Comparable shifts are also implicated by the change from Proto-DE *H-H and *H-L to Epie-Atissa L-HM, and from Proto-SW *H-H to Eruwa and Uvwie [evh] L-H. One could even understand the Degema change from *H-L to H-HL as of this type, albeit as spreading rather than shifting. In general, proto *H-H patterns appear very unstable across Edoid, no doubt tied to an Obligatory Contour Principle (OCP – Leben 1973) disfavoring identical tone sequences.

The last major type of change is found in South-West Edoid, where Proto-SW *L-LH changed in all languages: to L-H in Eruwa and Okpe but to L-L in the others. Rising contours within stems is marked in Edoid (clearly dispreferred and in fact nearly absent in DE and NW), and are entirely eliminated after L in SW by expanding either the low or high portion. We attribute this change to a type of constraint we call 'Rising-Markedness', where a rising sequence in a stem is dispreferred after a low prefix. Rising-Markedness involves perhaps the most abstract reconstructions and change type, since none of the descendent languages maintain the original *L-LH pattern.

5 Reconstructing Proto-Edoid tone

We can now present our reconstruction of Proto-Edoid (PE) tone using these intermediate reconstructions. As a first pass, we can plot the reconstructed values against each other to see how often they correspond. Table (36) below demonstrates this with the four melodies of Delta Edoid

(DE), the five of South West Edoid (SW), the five of North Central Edoid (NC), and the two of North West Edoid (NW). Cells where there is a disproportionate number of correspondences between two proto-melodies are in various shades of orange (where darker = higher number). In contrast, cells which appear to 3 or fewer cognates are in grey background (where darker = lower number).

(36) Cross-branch correspondence of intermediate reconstructed tone melodies

		SW					NC					NW	
		*L-L (*n*=91)	*L-LH (*n*=38)	*L-H (*n*=48)	*H-L (*n*=78)	*H-H (*n*=21)	*L-L (*n*=121)	*H-L (*n*=119)	*H-LH (*n*=53)	*H-H (*n*=27)	*L-H (*n*=27)	*L-L (*n*=56)	*H-L (*n*=103)
DE	*L-L (*n*=125)	**32**	**25**	11	16	3	**41**	**44**	9	2	7	**25**	**36**
	*H-H (*n*=39)	6	**1**	2	11	8	3	13	6	2	**1**	2	16
	*H-L (*n*=19)	**1**	**0**	4	4	2	2	4	3	4	**0**	1	6
	*L-H (*n*=19)	2	**1**	3	5	**0**	5	5	**1**	3	**0**	2	6
SW	*L-L (*n*=91)						**45**	14	7	5	4	**23**	13
	*L-LH (*n*=38)						3	**28**	2	**0**	**1**	**0**	**24**
	*L-H (*n*=48)						5	19	7	**1**	5	**1**	10
	*H-L (*n*=78)						**21**	15	19	8	4	12	16
	*H-H (*n*=21)						4	7	**0**	5	**1**	3	9
NC	*L-L (*n*=121)											**41**	5
	*H-L (*n*=119)											6	**60**
	*H-LH (*n*=53)											**1**	11
	*H-H (*n*=27)											3	9
	*L-H (*n*=27)											3	2

In order to account for this distribution in (36), we systematically compared the intermediate representations across the four branches in the same way we compared individual languages within a single branch. We are able to determine eight correspondence sets, which we label neutrally as classes PE-01 to PE-08, in (37) below. All together, these constitute 312 of the total 436 cognates (71.6%). We reconstructed the eight sets as a two way H/L contrast on the noun class prefix (PE-01 to PE-04 vs. PE-05 to PE-08), and a distinction between H/L/HL/LH on stems (recall that here and throughout, two ** mean PE). The number of members in each set is quite uneven, ranging from over a quarter of the cognates for PE-01 **L-L (*n*=113) down to only 3% for PE-04 **L-HL (*n*=13). In addition, there are a number of cognates which are ambiguous as to which set they belong to (PE+, labeled ‘ambiguous’ – *n*=88), as well as many which are simply irregular and not able to be placed within a set (PE-00, labeled ‘other’ – *n*=36). There is a fairly even distribution between cognates with ‘high’ confidence (matching a pre-established consonant correspondence) and those with ‘medium’ confidence (having a plausible but non-established consonant correspondence). Notably, in the four most common proto-melodies – **L-L, **L-H, **H-LH, and **H-H – there are more ‘high’ than ‘medium’ members. Also recall that cognates designed with ‘low’ confidence’ are those with only reflexes in a single branch, and are largely in the ‘ambiguous’ category.

(37) Proto-Edoid (PE) reconstruction

	PE-set	PE-reconstruction	*n*=	(	PE-confidence High	Med.	Low	)	Percentage of total (*n*=436)
a.	PE-01	**L-L	113	(	62	37	14	)	25.9%
b.	PE-02	**L-LH	27	(	9	14	4	)	6.2%
c.	PE-03	**L-H	38	(	21	15	2	)	8.7%
d.	PE-04	**L-HL	13	(	7	6	0	)	3.0%
e.	PE-05	**H-L	20	(	7	13	0	)	4.6%
f.	PE-06	**H-LH	54	(	26	21	7	)	12.4%
g.	PE-07	**H-HL	15	(	5	10	0	)	3.4%
h.	PE-08	**H-H	32	(	16	11	5	)	7.3%
i.	PE+	Ambiguous	88	(	25	26	37	)	20.2%
j.	PE-00	Other	36	(	12	6	18	)	8.3%

We based these Proto-Edoid values on their systematic reflexes in the intermediate reconstructions of the four Edoid branches. The PE melodies and their reflexes are in (38), grouped according to where there have been melody mergers. At the geographic extremes of Edoid (Map in (3)), P-DE and P-NW each have drastically reduced the Proto-Edoid system, with the resultant tone pattern primarily determined by the tone value of the prefix in Proto-Edoid. In contrast, P-

SW and P-NC in the interior of the Edoid spread zone are more conservative while still also showing class mergers largely based around whether there is a single H in the word or two H's (one on the prefix and one on the stem).

(38) Summary of PE proposal with reflexes in intermediate branches (mergers are boxed)

	PE		>	P-DE	P-SW	P-NC	P-NW
a.	**L-L	(PE-01)	>	*L-L	*…-L	*L-L	*L-L
b.	**L-LH	(PE-02)	>	*L-L	*L-H	*L-H	*L-L
c.	**H-L	(PE-05)	>	*L-L	*L-H	*H-L	*H-L
d.	**L-H	(PE-03)	>	*L-L	*L-LH	*H-L	*H-L
e.	**L-HL	(PE-04)	>	?	*L-L	*H-L	*H-L
f.	**H-HL	(PE-07)	>	*H-…	*H-L	*H-L	*H-L
g.	**H-LH	(PE-06)	>	*H-…	*H-L	*H-LH	*H-L
h.	**H-H	(PE-08)	>	*H-…	*H-…	*H-…	*H-L

In what follows in this section, we expand on each of these eight PE correspondence sets, first discussing **L-L and **L-LH with uniform L-initial reflexes (the 'low' grouping – §5.1), then **H-L, **L-H, and **L-HL with both H- and L-initial reflexes (the 'high/low' grouping – §5.2), and finally **H-HL, **H-LH, and **H-H with only H-initial reflexes (the 'high' grouping – §5.3).

5.1 The 'low' grouping

The 'low' grouping consists of two correspondence sets, one reconstructed as PE **L-L (PE-01) and one as **L-LH (PE-02). The former set is by far the largest in PE (n=113), twice the size of the next largest set. Just as we reconstructed major and minor subsets for each correspondence set for the four individual Edoid branches, we may also do so at the PE level, shown in (39). The majority of **L-L reflexes show *L-L across the four branches (as in PE-01A), but a notable minority show *H-L in P-SW (PE-01B). Others are ambiguous because a P-SW form is incidentally missing (PE-01U), and still others show other irregularities such as a high in some other position or language (PE-01X).

(39) Reflexes of PE-01 **L-L in four daughter branches (*n*=113)

Subset	*n*=	Cognate	P-DE	P-SW	P-NC	P-NW
PE-01A	46	TOOTH	*L-L	*L-L	*L-L	*L-L
PE-01B	18	ASH	*L-L	*H-L	*L-L	*L-L
PE-01U	31	DOOR_2	*L-L	-	*L-L	*L-L
PE-01X	18	SOUP				

Example reflexes in present-day Edoid languages are shown in (40)-(41). Table (40) shows two PE-01A cognates TOOTH and FOOT_LEG_2, which show the expected reflexes in each language for *L-L (recall H-Obligatoriness changes as discussed in §4.5 most recently).

(40) PE-01A **L-L cognates TOOTH and FOOT_LEG_2 (with *L-L reflexes)

	Branch	Language	TOOTH	Tones	Proto-T	FOOT_LEG_2	Tones	Proto-T
a.	DE	deg	**ʊ̀-kʊ́**	L-H	*L-L	**à-βí**	L-H	*L-L
		enn	**à-kʊ̀**	L-L		**à-vì**	L-L	
		epi	**à-kʊ̏**	L-L		**à-vì**	L-L	
b.	SW	erh	**à-kɔ́**	L-H	*L-L	**à-ʋɔ̀**	L-L	*L-L
		iso	**à-kɔ̀**	L-L	(SW-01A)	**ʊ̀-wɔ̀**	L-L	(SW-01C)
		oke	**á-wȍ**	H-L		**ò-wɔ̀**	L-L	
		urh	**à-kȍ**	L-L		**ò-wɔ̀**	L-L	
		evh	**à-kʊ̏**	L-L		**à-ʋɔ̀**	L-L	
c.	NC	bin	**à-kɔ̏**	L-L	*L-L	**ò-wɛ̀**	L-L	*L-L
		ish	**à-kɔ̏**	L-L		**ò-wɛ̀**	L-L	
		ema	**à-kɔ̏**	L-L		**ò-ɛ̀**	L-L	
		ets	**ɛ̀-kɔ̀**	L-L		**ò-wɛ̀**	L-L	
		une	**à-kɔ̀**	L-L		**-**	-	
		aaa	**ɛ̀ɛ̀-kɔ̀**	L-L		**-**	-	
		oso	**[í]rɛ̀-kɔ̀**	[H]L-L		**ò-wɛ̀**	L-L	
d.	NW	atg	**à-kò**	L-L	*L-L	**ò-wɛ̀**	L-L	*L-L
		olm	**lɛ̂-kɔ̀**	F-L		**ɣɔ̂-mhì[nà]**	F-L[L]	
		opa	**rà-kɔ̀**	L-L		**-**	-	
		uha	**à-kɔ̀**	L-L		**-**	-	
		ehu	**à-kȕ**	L-L		**-**	-	
		uku	**à-kȕ**	L-L		**-**	-	

In contrast, (41) shows PE-01B cognates ASH and FAECES_1, which have a *H-L in P-SW (in red), and consequently have an initial H- in all SW daughter languages. At this point, there is no obvious conditioning factor between PE **L-L becoming P-SW *L-L vs. *H-L. We shall interpret this as a variable application of H-Obligatoriness in the development of PE to P-SW.

(41) PE-01B **L-L cognates ASH and FAECES_1 (with exceptional *H-L reflex in P-SW)

	Branch	Language	ASH	Tones	Proto-T	FAECES_1	Tones	Proto-T
a.	DE	deg	**ə-mə[fú]**	L-L[H]	*L-L	**ì-sɔ́**	L-H	*L-L
		enn	**è-ɓù[fù]**	L-L[L]		-	-	
		epi	**ù-wò**	L-L		-	-	
b.	SW	erh	**í-mʊ̀**	H-L	*H-L	-	-	*H-L
		iso	**é-ŋwò**	H-L	(SW-04A)	**í-sɔ̀**	H-L	(SW-04B)
		oke	-	-		**í-sɔ̃̀**	H-L	
		urh	**í-wù[r̥iè]**	H-L[L]		**í-ꜜsɔ̃́**	H-ꜜH	
		evh	-	-		**í-sɔ̃̀**	H-L	
c.	NC	bin	**è-ʊ̃ɛ̃̀**	L-L	*L-L	**ì-sã̀**	L-L	*L-L
		ish	**é-ʋɔ̃̀**	H-L		-	-	
		ema	**è-mùɛ̀**	L-LL		**ì-sɔ̃̀**	L-L	
		ets	**è-ŋwɛ̀ɛ̀**	L-LL		**ì-tsɔ̀**	L-L	
		une	**è-mhɔ̀**	L-L		**ì-sɔ̀**	L-L	
		aaa	**ɣè-mhɔ̀**	L-L		**ì-sɔ̀**	L-L	
		oso	**è-mwɛ̂**	L-F		**ì-sɔ̀**	L-L	
d.	NW	atg	**è-nwɛ̀**	L-L	*L-L	-	-	*L-L
		olm	**ghɛ̂-mɔ̀**	F-L		**ì-sɔ̀**	L-L	
		opa	**wɛ̀-mɔ̀**	L-L		**ì-sò**	L-L	
		uha	**ɛ̀-mɔ̀**	L-L		**í-sɔ̀**	H-L	
		ehu	**è-mɔ̀**	L-L		**ì-sɔ̃̀**	L-L	
		uku	**è-mɔ̀**	L-L		**ì-ɹ̃ɔ̃̀**	L-L	

The next correspondence set is PE-02 **L-LH (*n*=27), whose reflex is *L-H in the more centrally located P-SW and P-NC, but *L-L in the more peripheral P-DE and P-NW branches (42). The expected reflexes are subset PE-02A, such as WALKING_STICK expanded in (43) with modern reflexes. Additional subsets include those with an unexpected *L-L reflex in P-SW (PE-02B, e.g. BLOOD_2) or *L-L in P-NC (PE-02C). As apparent from the example cognates in (43), this correspondence set is marked by fewer reflex forms in individual languages, reducing our confidence in its interpretation. As a whole, **L-LH is dwarfed by the number of **L-L cognates (more than four times the size), and the majority of *L-LH cognates are of medium confidence while the majority of *L-L are of high confidence (see (37)).

(42) Reflexes of PE-02 **L-LH (n=27)

Subset	n=	Cognate	P-DE	P-SW	P-NC	P-NW
PE-02A	11	WALKING_STICK	*L-L	*L-H	*L-H	*L-L
PE-02B	4	BLOOD_2	*L-L	*L-L	*L-H	*L-L
PE-02C	3	PROVERB	*L-L	*L-H	*L-L	*L-L
PE-02U	4	COWIFE_1	-	-	*L-H	-
PE-02X	5	ELDER_4				

(43) Reflexes of PE **L-LH – WALKING_STICK (PE-02A) and BLOOD_2 (PE-02B)

Branch	Language	WALKING_STICK	Tones	Proto-T	BLOOD_2	Tones	Proto-T
a. DE	deg	**ò-kpòtí**	L-LH	*L-L	**ì-zàlà**	L-LL	*L-L
	enn	-	-		-	-	
	epi	-	-		**ì-zàrà**	L-LL	
b. SW	erh	-	-	*L-**H**	**à-zí**	L-H	*L-**L**
	iso	-	-		**à-zɛ̀**	L-L	
	oke	-	-		-	-	
	urh	**ɔ̀-kpɔ́**	L-H		-	-	
	evh	-	-		-	-	
c. NC	bin	**ɔ̀-kpá**	L-H	*L-H	**è-ságjɛ̂**	L-HL	*L-H
	ish	**[úkp]ò-kpó**	[H]L-H		-	-	
	ema	**ɔ́-ꜜkpɔ́**	H-ꜜH		**è-sádʒɛ̂**	L-HL	
	ets	-	-		-	-	
	une	-	-		-	-	
	aaa	-	-		**ò-òdɛ̀**	L-LL	
	oso	-	-		**ò-zɛ̀**	L-L	
d. NW	atg	-	-		-	-	*L-L
	olm	-	-		**â-zɛ̀**	F-L	
	opa	-	-		**à-dzɛ̀**	L-L	
	uha	-	-		**ɛ̀-zɛ̀**	L-L	
	ehu	-	-		**ɛ̀-rɛ̀**	L-L	
	uku	-	-		**è-dɛ̀**	L-L	

The change from PE **L-LH to *L-L or *L-H (with variability in some cognates) directly mirrors the behavior of *L-LH in SW we saw in §4.2, whose reflex was L-L or L-H depending on the language. We turn next to the origin of SW *L-LH, where we analyze it as from PE **H-L.

5.2 The ‘high/low’ grouping

Next is the ‘high/low’ grouping, consisting of three correspondence sets: **L-H (PE-03), **L-HL (PE-04), and **H-L (PE-05). These three have in common that they show either a H- or L- prefix in the four descendent proto-languages, whereas the other groupings show a consistent initial L- or initial H- reflex. These three proto-melodies share the fact that they have an opposite value at the internal stem boundary, which will become relevant to understanding the total changes which took place in the development of Edoid tone.

Starting with PE-03 **L-H (*n*=38), this becomes *L-L in P-DE, *L-LH in P-SW, and *H-L in P-NC and P-NW, shown in (44). Two cognates are provided in (45), one with the expected reflex (HEAD – PE03A) and one with some irregularity (in this example concentrated within the DE branch for HOUSE_1 – PE03X).

(44) Reflexes of PE-03 **L-H (*n*=38)

Subset	*n*=	Cognate	P-DE	P-SW	P-NC	P-NW
PE-03A	24	HEAD	*L-L	*L-LH	*H-L	*H-L
PE-03B	2	URINE_1	*L-L	*L-L	*H-L	*H-L
PE-03U	2	GIRL_2		-		
PE-03X	10	HOUSE_1				

(45) Reflexes of PE **L-H – HEAD (PE-03A) and HOUSE_1 (PE-03X)

	Branch	Language	HEAD	Tones	Proto-T	HOUSE_1	Tones	Proto-T
a.	DE	deg	**ù-tóm**	L-H	*L-L	**ʊ́-ꜜβáj**	H-ꜜH	*H-H
		enn	**ù-tòmù**	L-LL		**-**	-	
		epi	**ù-tòm**	L-L		**ɔ̀-wɔ̀ɔ̀**	L-LL	
b.	SW	erh	**ù-zòʊʊ̋**	L-LH	*L-LH	**ʊ̀-wá**	L-H	*L-LH
		iso	**ù-zòù**	L-LL		**-**	-	
		oke	**ù-ɽ̥ómú**	L-HH		**ò-ɣwá**	L-H	
		urh	**ù-jȍʊȉ**	L-LL		**ù-wèʊì**	L-LL	
		evh	**-**	-		**ù-ɣwòmù**	L-LL	
c.	NC	bin	**ù-hűɱȕ**	L-HL	*H-L	**ò-wá**	L-H	*H-L
		ish	**ú-hɔ̏βɔ̏**	H-LL		**ú-wâ**	H-F	
		ema	**ú-hùmȉ**	H-LL		**ó-à**	H-L	
		ets	**ú-sò**	H-L		**ó-ʋà**	H-L	
		une	**ú-sòmhì**	H-LL		**ó-ʋà**	H-L	
		aaa	**ū-sɔ̀**	M-L		**ō-ʋàɣì**	M-LL	
		oso	**ú-kwɛ̀**	H-L		**ó-wà**	H-L	
d.	NW	atg	**ú-xòmhì**	H-LL	*H-L	**ó-wà**	H-L	*H-L
		olm	**ɣí-kà**	H-L		**-**	-	
		opa	**ú-xèmhì**	H-LL		**-**	-	
		uha	**ú-kèmì**	H-LL		**-**	-	
		ehu	**ū-hȕ**	M-L		**-**	-	
		uku	**ú-kɔ̀mì**	H-LL		**-**	-	

For these melodies, PE is reconstructed with an initial L- (in **L-H) and the P-NC and P-NW branches show an initial H-. We must therefore posit an externalization process moving stem tone to the prefix position. This process will be seen several more times as we proceed, and will be discussed in more detail in §5.4.

Next consider PE-04 reconstructed as **L-HL in (46). This has a *H-L reflex in both P-NC and P-NW, but unlike what we saw above here P-SW is *L-L. Interestingly, P-DE shows variation between *L-L (PE-04A) and *H-H reflexes (PE-04B), though one should note the very low number of members in this correspondence set (*n*=13). This undercuts our certainty in these patterns, especially what to take as the expected reflexes in P-DE and P-SW.

(46) Reflexes of PE-04 **L-HL (*n*=14)

Subset	*n*=	Cognate	P-DE	P-SW	P-NC	P-NW
PE-04A	4	GREY_HAIR	*L-L	*L-L	*H-L	*H-L
PE-04B	3	DOG_1	*H-H	*L-L	*H-L	*H-L
PE-04U	3	NEST	-			
PE-04X	3	SONG_2				

(47) Reflexes of PE **L-HL – GREY_HAIR (PE-04A) and DOG_1 (PE-04B)

	Branch	Language	GREY_HAIR	Tones	Proto-T	DOG_1	Tones	Proto-T
a.	DE	deg	**ʊ̀-dáɲ**	L-H	*L-L	**á-ꜜɓʊ́wá**	H-ꜜHH	*H-H
		enn	**ɛ̀-dàà**	L-LL		**á-ɓʊ̀à**	H-LL	
		epi	**ì-dàn**	L-L		**à-ɓʊ́ā**	L-HM	
b.	SW	erh	**-**	-	*L-L	**-**	-	*L-L
		iso	**-**	-		**ɛ̀-bɔ̀**	L-L	
		oke	**é-r̥à**	H-L		**-**	-	
		urh	**è-ʃà**	L-L		**-**	-	
		evh	**-**	-		**-**	-	
c.	NC	bin	**è-dɛ́**	L-H	*H-L	**à-wá**	L-H	*H-L
		ish	**-**	-		**á-wà**	H-L	
		ema	**é-dɛ̀**	H-L		**á-wà**	H-L	
		ets	**é-dɛ̀**	H-L		**á-ɣwà**	H-L	
		une	**é-dɛ̀**	H-L		**á-wà**	H-L	
		aaa	**ghō-dɛ̀**	M-L		**ɣā-wà**	M-L	
		oso	**-**	-		**á-wà**	H-L	
d.	NW	atg	**é-dɛ̀**	H-L	*H-L	**à-wó[ʃí]**	L-H[H]	*H-L
		olm	**ghó-zɛ̀**	H-L		**ghá-wà**	H-L	
		opa	**-**	-		**wá-bʊ̀à**	H-LL	
		uha	**-**	-		**à-bùà**	L-LL	
		ehu	**-**	-		**ɛ̄-bɔ̀**	M-L	
		uku	**-**	-		**ɛ́-bɔ̀**	H-L	

The final 'high/low' correspondence set is PE-05 reconstructed as **H-L (48). In this set, P-NC and P-NW again become *H-L, while P-SW becomes *L-H and P-DE becomes *L-L (for the most part).

(48) Reflexes of PE-05 **H-L (*n*=20)

Subset	*n*=	Cognate	P-DE	P-SW	P-NC	P-NW
PE-05A	12	TORTOISE_1	*L-L	*L-H	*H-L	*H-L
PE-05X	8	ANTELOPE_1				

(49) Reflexes of PE **H-L – TORTOISE_1 (PE-05A) and ANTELOPE_1 (PE-05X)

Branch	Language	TORTOISE_1	Tones	Proto-T	ANTELOPE_1	Tones	Proto-T
a. DE	deg	**è-wíj**	L-H	*L-L	**é-túwèj**	H-HL	*H-L
	enn	**è-wìì**	L-LL		**é-tùèì**	H-LLL	
	epi	**è-wìì**	L-L		**ò-túêì**	L-HFL	
b. SW	erh	**-**	-	*L-H	**è-rúé**	L-HH	*L-H
	iso	**ò-gbé**	L-H		**è-rúɛ́**	L-HH	
	oke	**-**	-		**è-ɽ̥úɛ́**	L-HH	
	urh	**ò-gbéjĩ̄**	L-HM		**è-ɽ̥ṹɛ̃́**	L-HH	
	evh	**-**	-		**è-ɽ̥ṹɛ̃́**	L-HH	
c. NC	bin	**è-gúí**	L-HH	*H-L	**ɛ̀-ɽ̥ùɛ̀**	L-LL	*H-L
	ish	**é-wì**	H-L		**ɛ́-rùɛ̀**	H-LL	
	ema	**é-ĩ̀**	H-L		**-**	-	
	ets	**é-ɣì**	H-L		**ɛ́-ɽ̥wè**	H-L	
	une	**é-hũ̀**	H-L		**ɛ́-θwè**	H-L	
	aaa	**ē-ũ̀**	M-L		**ɛ̄-ɽ̥ùɛ̀**	M-LL	
	oso	**é-gù**	H-L		**-**	-	
d. NW	atg	**-**	-	*H-L	**-**	-	*H-L
	olm	**-**	-		**á-tù**	H-L	
	opa	**é-gùì**	H-LL		**-**	-	
	uha	**é-gì**	H-L		**-**	-	
	ehu	**ē-gùè**	M-LL		**ɛ̀-ɽ̥ũ̀ɛ̃̀**	L-LL	
	uku	**é-gùì**	H-LL		**ɛ́-tũ̀ɛ̃̀**	H-LL	

The crucial aspect of the PE-05 set is that while the **H-L value remains as such in P-NC and P-NW, there is a rightward shift of the H tone in P-SW. This is analogous to other rightward shifts already seen, e.g. the Great Edo Tone Shift (§4.3). We expand on the various shifts in PE in §5.4 and §6.1 below.

5.3 The 'high' grouping

Lastly, the 'high' grouping consists of three correspondence sets whose reflexes show an initial H- in all four intermediate proto-languages. These are **H-LH (PE-06), **H-HL (PE-07), and

**H-H (PE-08), which have in common that they have a H both on the prefix and somewhere within the stem. Before we detail these sets, we must admit that this grouping is perhaps the most tentative with frequent subsets of unclear conditioning. This is true as a whole of the Delta branch which corresponds to P-DE *H-L or *H-H in each of the three sets, and particularly characteristic of PE-08 **H-H which shows noticeable irregularity.

Beginning with PE-06, this is reconstructed as **H-LH. This remains as *H-LH in P-NC and becomes *H-L in P-NW (50). It becomes either *H-L or *H-H in P-DE, with too much variation to privilege one over the other. It is in P-SW that we see the most significant variation. While most P-SW reflexes of this set are *H-L (PE-06A), a minority are *L-H (PE-06B) or *L-L (PE-06C). Still others are either ambiguous in its subset (PE-06U) or cannot be fit into these subsets (PE-06X).

(50) Reflexes of PE-06 **H-LH (*n*=54)

Subset	*n*=	Cognate	P-DE	P-SW	P-NC	P-NW
PE-06A	14	SNAIL	*H-L/*H-H	*H-L	*H-LH	*H-L
PE-06B	5	GRASSCUTTER_3	*H-L/*H-H	*L-H	*H-LH	*H-L
PE-06C	4	BOUNDARY_1	*H-L/*H-H	*L-L	*H-LH	*H-L
PE-06U	12	COLD_N		-		
PE-06X	19	BIRD				

(51) Reflexes of PE **H-LH – SNAIL (PE-06A) and BOUNDARY_1 (PE-06C)

Branch	Language	SNAIL	Tones	Proto-T	BOUNDARY_1	Tones	Proto-T
a. DE	deg	**-**	-		**ó-ꜜgúrúrú**	H-ꜜHHH	*H-H
	enn	**-**	-		**-**	-	
	epi	**-**	-		**-**	-	
b. SW	erh	**-**	-	*H-L	**-**	-	*L-L
	iso	**ù-sè[kpè]**	L-L[L]		**-**	-	
	oke	**-**	-		**-**	-	
	urh	**ú-sè[kpè]**	H-L[L]		**ù-ɣwrù**	L-L	
	evh	**-**	-		**-**	-	
c. NC	bin	**ú-r̥èé**	H-LH	*H-LH	**ú-ɣù**	H-L	*H-LH
	ish	**ú-ꜜrê**	H-ꜜF		**ú-ɣù**	H-L	
	ema	**ú-ꜜré**	H-ꜜH		**-**	-	
	ets	**ú-r̥ě**	H-R		**-**	-	
	une	**ú-θèì**	H-LL		**-**	-	
	aaa	**ū-ré**	M-H		**-**	-	
	oso	**-**	-		**-**	-	
d. NW	atg	**ú-lùé**	H-LH	*H-L	**-**	-	
	olm	**ú-thù[nhámhì]**	H-L[HL]		**-**	-	
	opa	**-**	-		**-**	-	
	uha	**-**	-		**-**	-	
	ehu	**-**	-		**-**	-	
	uku	**-**	-		**-**	-	

Furthermore, the much smaller PE-07 is reconstructed as **H-HL. P-DE and P-NW branches have identical reflexes as with PE-06 **H-LH above, namely *H-L/*H-H and *H-L respectively. The key differences are in P-SW which consistently shows a *H-L pattern and P-NC which shows *H-L rather than *H-LH. This is evidenced most clearly by the fact that reflexes of P-NC *H-L (< PE **H-HL) in Edo undergo the Great Edo Tone Shift, while reflexes of P-NC *H-LH (< PE **H-LH) do not.

(52) Reflexes of PE-07 **H-HL (*n*=15)

Subset	*n*=	Cognate	P-DE	P-SW	P-NC	P-NW
PE-07A	9	RIVER	*H-L/*H-H	*H-L	*H-L	*H-L
PE-07X	6	ROPE_1				

(53) Reflexes of PE **H-HL – RIVER (PE-07A) and ROPE_1 (PE-07X)

	Branch	Language	RIVER	Tones	Proto-T	ROPE_1	Tones	Proto-T
a.	DE	deg	**é-ꜜdə́**	H-ꜜH	*H-H	**ù-ɗíŋ^{w}**	L-H	*L-L
		enn	**é-dà**	H-L		**ì-ɗìì**	L-LL	
		epi	**é-dē**	H-M		**ù-ɗìì**	L-LL	
b.	SW	erh	**í-sɛ̀**	H-L	*H-L	-	-	*H-L
		iso	**é-t̪ɛ̀**	H-L		-	-	
		oke	-	-		-	-	
		urh	-	-		**ú-rùr̥ɛ́**	H-LH	
		evh	-	-		-	-	
c.	NC	bin	**ɛ̀-zɛ̀**	L-L	*H-L	**ì-rí**	L-H	*H-L
		ish	**ɛ́-dɛ̀**	H-L		**[úkp]í-rì**	[H]H-L	
		ema	**ɛ́-dà**	H-L		**ú-ì**	H-L	
		ets	**ɛ́-dà**	H-L		**ú-lì**	H-L	
		une	**ɛ́-dɔ̀**	H-L		**ú-lì**	H-L	
		aaa	**ɛ̄-dà**	M-L		**ū-ùɣì**	M-LL	
		oso	-	-		**ú-rídìà**	H-HL	
d.	NW	atg	-	-	*H-L	-	-	*H-L
		olm	-	-		**ghú-rìghì**	H-LL	
		opa	-	-		**ú-rì**	H-L	
		uha	-	-		**ú-dì**	H-L	
		ehu	-	-		**ù-rī**	L-M	
		uku	-	-		-	-	

As with PE-06 **H-LH, the prefix H- in the PE-07 **H-HL melody does not shift rightward in P-SW (hence the reconstructed value *H-L for both RIVER and ROPE_1 in (53)). This contrasts with PE-05 **H-L, where it does shift into the stem in P-SW (resulting in *L-H – see (49) in the previous subsection). The key difference is that the former melodies have a stem H which prevents H-tone shift whereas the latter **H-L melody does not.

The last set is PE-08 reconstructed as **H-H. This has the most subsets and exhibits frequent variation, which may suggest that it actually constitutes more than one proto-melody which is no longer distinguishable (at least not at this time). One impetus for the **H-H set in PE is to account for the presence of H-H reflexes in *SW and *NC which have not yet been shown to be reflexes of any PE category. The largest subset is PE-08A, which has a *H-H reflex in P-SW, *H-L in P-

NC and P-NW, and variably *H-H/*H-L in P-DE. At the same time, there are also a few examples of P-NC *H-H which corresponds either to P-SW *H-H (PE-08B) or *H-L (PE-08C).

(54) Reflexes of PE-08 **H-H (*n*=32)

Subset	*n*=	Cognate	P-DE	P-SW	P-NC	P-NW
PE-08A	6	EAR	*H-L/*H-H	*H-H	*H-L	*H-L
PE-08B	2	NOSE	*H-L/*H-H	*H-H	*H-H	*H-L
PE-08C	3	STALL	*H-L/*H-H	*H-L	*H-H	*H-L
PE-08U	9					
PE-08X	12					

(55) Reflexes of PE **H-H – EAR and TREE (PE-08A)

	Branch	Language	EAR	Tones	Proto-T	TREE	Tones	Proto-T
a.	DE	deg	**ʊ́-ꜜsɔ́**	H-ꜜH	*H-H	**ʊ́-ꜜtáɲ**	H-ꜜH	*H-H
		enn	**ɛ́-sɔ̀ʊ̀**	H-LL		**ɛ́-tàì**	H-LL	
		epi	**í-sɔ́ɔ̄**	H-HM		**ì-táān**	L-HM	
b.	SW	erh	**-**	-	*H-H	**ʊ́-rã̀rɛ̃̀**	H-LL	*H-H
		iso	**ʊ́-zɔ́**	H-H		**ú-ré**	H-H	
		oke	**ò-r̥ɔ́**	L-H		**ò-r̥ã́**	L-H	
		urh	**ó-r̥ɔ́**	H-H		**ú-r̥é**	H-H	
		evh	**è-sɔ̃́**	L-H		**ʊ̀-rá**	L-H	
c.	NC	bin	**è-hɔ́**	L-H	*H-L	**è-r̥ã́**	L-H	*H-L
		ish	**é-hɔ̀**	H-L		**é-rã̀**	H-L	
		ema	**é-hɔ̃̀**	H-L		**ó-rã̀**	H-L	
		ets	**é-ʋɔ̀**	H-L		**ó-r̥à**	H-L	
		une	**é-sɔ̀**	H-L		**ó-θà**	H-L	
		aaa	**ɣō-hɔ̀ɣì**	M-LL		**ō-tà**	M-L	
		oso	**ó-sɔ̀**	H-L		**ó-thɛ̀**	H-L	
d.	NW	atg	**é-sɔ̀**	H-L	*H-L	**ó-rɛ̀**	H-L	*H-L
		olm	**ghɔ́-z̈ɔ̀**	H-L		**ɔ́-shà**	H-L	
		opa	**wʊ́-z̈ɔ̀**	H-L		**ʊ́-ʃà**	H-L	
		uha	**ɛ́-sɔ̀**	H-L		**ó-ʃà**	H-L	
		ehu	**é-r̥ɔ̀**	H-L		**è-ʃā**	L-M	
		uku	**é-r̥ɔ̀**	H-L		**ò-ʃâ**	L-F	

(56) Reflexes of PE **H-H – NOSE (PE-08B) and STALL (PE-08C)

	Branch	Language	NOSE	Tones	Proto-T	STALL	Tones	Proto-T
a.	DE	deg	**í-súβèɲ**	H-HL	*H-L	-	-	
		enn	**ú-swèì**	H-LL		-	-	
		epi	**ù-súɓē**	L-HM		-	-	
b.	SW	erh	**û-wɛ̀**	F-L	*H-H	-	-	*H-L
		iso	**ú-ŋwé**	H-H		-	-	
		oke	**ì-r̥úɛ́**	L-HH		-	-	
		urh	**ú-wɛ́**	H-H		**ó-ɣwà**	H-L	
		evh	-	-		-	-	
c.	NC	bin	**í-húé**	H-HH	*H-H	**ó-wá**	H-H	*H-H
		ish	**í-ꜜhwê**	H-ꜜF		**ó-wáà**	H-HL	
		ema	**í-hùè**	H-LL		-	-	
		ets	**í-sùè**	H-LL		**ò-gwá[kì]**	L-H[L]	
		une	**í-ʃùè**	H-LL		**é-gbhá[kò]**	H-H[L]	
		aaa	**ī-rùè**	M-LL		-	-	
		oso	**í-fwè**	H-L		-	-	
d.	NW	atg	-	-	*H-L	-	-	
		olm	**ú-žùò**	H-LL		-	-	
		opa	**ú-sùò**	H-LL		-	-	
		uha	**í-sù**	H-L		-	-	
		ehu	**í-r̥ò**	H-L		-	-	
		uku	**í-r̥ò**	H-L		-	-	

5.4 Summary of proposed tone changes from Proto-Edoid

To summarize, we have posited eight tonal correspondence sets for proto-Edoid, divided into three groupings (57). A ‘low’ grouping consists of **L-L and **L-LH melodies, a ‘high/low’ grouping consists of **L-H, **L-HL, and **H-L melodies, and a high grouping consists of **H-LH, **H-HL, and **H-H melodies. In the summary table in (57), we notate any changes from Proto-Edoid to the intermediate reconstructions in red, and box any mergers.

(57) Proto-Edoid and reflexes in four intermediate branches (changes in red, mergers boxed)

		Proto-Edoid			Proto-DE	Proto-SW	Proto-NC	Proto-NW
a.	‘Low’	PE-01	**L-L	>	*L-L	*L-L/*H-L	*L-L	*L-L
	grouping	PE-02	**L-LH	>	*L-L	*L-H	*L-H	*L-L
b.	‘High/low’	PE-05	**H-L	>	*L-L	*L-H	*H-L	*H-L
	grouping	PE-03	**L-H	>	*L-L	*L-LH	*H-L	*H-L
		PE-04	**L-HL	>	?	*L-L	*H-L	*H-L
c.	‘High’	PE-07	**H-HL	>	*H-H/*H-L	*H-L	*H-L	*H-L
	grouping	PE-06	**H-LH	>	*H-H/*H-L	*H-L	*H-LH	*H-L
		PE-08	**H-H	>	*H-H/*H-L	*H-H/*H-L	*H-L/*H-H	*H-L

Looking across the four branches, we notice largely the same array of diachronic tone changes that we established for the four intermediate branches to modern Edoid languages (§4.5 above). H-Obligatoriness in Proto-Edoid – the addition of a high tone to all-low sequences – is not discernable to the same degree as it is for individual Edoid languages, though we still see some effects of it (PE **L-L > P-SW *H-L, albeit with an unclear conditioning environment). Another established change, H-Lowering, is more common and is found in all four branches in the ‘High’ grouping. Further, Rightward H-Shift happened in the development of P-DE and P-SW where the H of **H-L melodies moves into the stem (eventually lost in P-DE), which is blocked from happening if the stem already contains a H tone. This is directly analogous to the conditions of the Great Edo Tone Shift from P-NC *H-L to modern-day Edo L-H (§4.3). Finally, just as we saw rising sequences as marked in SW (called ‘Rising-Markedness’), **L-LH too is marked in PE and eliminated in favor of *L-L and *L-H.

While these diachronic changes do indeed recur from PE to P-DE/P-SW/P-NC/P-NW and then from these intermediate branches to the modern languages, there are additional diachronic changes which seem to have taken place only in PE. One notable tone change is a ‘Leftward H-Shift’ in P-NC and P-NW which changed PE **L-H and **L-HL into *H-L, and acts as the logical counterpart to the more common Rightward H-Shift. Specifically, Leftward H-Shift only takes place if the **H is at the stem boundary already – it is not found if it is further away, e.g. **L-LH becomes *L-H/*L-L (not *H-L). Typologically, leftward shift is much rarer than rightward shift, an observation we return to in the discussion section (§6.1).

Another set of new diachronic changes are implied by the development of P-SW, which retains much of the tonal contrast of PE but with key reorganizations. One is Low-Plateauing in which H is lowered between two L's, i.e. PE **L-HL > P-SW *L-L. More complicated is what appears to be swapping of two values: PE **L-H becomes P-SW *L-LH but PE **L-LH becomes P-SW *L-H. Whether this is the best analysis and how such a swap could come about (likely through a series of changes) remain for future work.

In total, all branches involve a reduction of the tone classes of PE, especially with respect to multiple tone values on the stem portion of the noun. P-DE collapses the distinction essentially into a *L-L vs. *H-H/*H-L distinction based on the original tone on the prefix (*modulo* the rightward shift of the H in **H-L first into the stem). A similar collapse happened in P-NW, where there is a *L-L vs. *H-L distinction also dependent on the tone on the prefix (again, *modulo* the leftward shift of H in **L-H and **L-HL to the prefix position). In contrast, P-SW and P-NC maintain more original contrasts but by different means. P-SW shifts H rightward in **H-L (but not with other H- initial contexts where the stem H blocks shift), while P-NC shifts the H of **L-LH, **L-H, and **L-HL leftward one position (but not with *H-LH due to the presence of the prefix H). In total, based on this reconstruction the most important tone preserved from PE is the prefix tone, followed by the stem-initial tone, and finally the stem-final tone.

6 Discussion

Having established our proposal for Proto-Edoid tone, we discuss four issues: interpreting Edoid within the larger typology of tone changes (§6.1), assessing the regularity of Edoid tone change

(§6.2), deliberating over whether SW Edoid is innovative or conservative (§6.3), and finally implications of Proto-Edoid for Proto-Benue-Congo and Proto-Niger-Congo (§6.4).

6.1 Edoid within the typology of diachronic tone changes

Within each stage in the development Edoid tone, we have posited a specific reconstructed value, even when this meant a reconstruction which was not maintained in any descendent language. For example, for the three P-SW correspondence sets SW-01, SW-02, and SW-03, in order to account for the distribution of L-L and L-H in the modern SW languages, we reconstructed these sets as *L-L, *L-LH, and *L-H, respectively, even though no language maintains the *L-LH pattern. This is opposed to an alternative which would simply reconstruct these as $*L_1$, $*L_2$, and $*L_3$ (or $*LH_1$, $*LH_2$, and $*LH_3$, or some combination).

Consequently, for each change in a descendent cognate compared to the proto-melody, we posit a precise diachronic tone change which is both typologically plausible and forms a coherent set of changes internal to the language. This is important for cross-linguistic work on tone reconstruction, because the typology of tone change sorely lags behind that of segment change, which reduces our collective knowledge on what is common vs. rare vs. unattested (for similar sentiments, see Campbell 2021).

In total, we have posited six types of tone changes when examining all stages in the development of Edoid, summarized in (58).

(58) Diachronic tone changes grouped into six types in all stages of Edoid development

a. Rightward H-Shift (e.g. H-L > L-H)

b. Leftward H-Shift (e.g. L-H > H-L)

c. H-Lowering (e.g. H > M; H-H > H-$^{\downarrow}$H; L-HL > L-L; *etc.*)

d. L-Raising (e.g. H-L > H-$^{\downarrow}$H or H-M)

e. H-Obligatoriness (e.g. L-L > H-L or L-H)

f. Rising-Markedness (e.g. L-LH > L-H or L-L)

The occurrence of these changes is uneven across the various stages of Edoid. This is summarized in (59), which divides PE to the four proto-branches, and then from these intermediate reconstructions to the modern-day languages. If a language demonstrates the tone change, its cell is marked with an 'X' (which we place in parentheses if the change applies to only a minority subset of the relevant cognates).

(59) Diachronic tone changes by individual language

	PE				DE			SW					NC							NW					
	P-DE	P-SW	P-NC	P-NW	deg	enn	epi	erh	iso	oke	urh	evh	bin	esa	ema	ets	une	aaa	oso	atg	olm	opa	uha	ehu	uku
R'wrd H-Shift (HL > LH)	X	X					X						X												
L'wrd H-Shift (LH > HL)			X	X																					
H-Lowering (H > $^{\downarrow}$H/M/L)	X	X	X	X	X	X	X	X		X		X		X	X	X	X	X	X					X	
L-Raising (L > $^{\downarrow}$H/M)							X				(X)														
H-Obligatory (∅ > H)		(X)			X			X		X			(X)								X				
Rise-Marked (-LH > -L/-H)	X	X	X	X				X	X	X	X	X													

Let us focus first on the top two involving H-shifting. From (59), leftward shift is only hypothesized for P-NC and P-NW to account for correspondences to P-DE and P-SW L-initial cognates. In contrast, evidence for rightward shift is clearly seen in modern day languages, such as Edo where P-NC *H-L becomes L-H systematically (the Great Edo Tone Shift – §4.3). The prominence of rightward over leftward shift complies with cross-linguistic trends. Perseverative H-shifting or H-spreading rightward is much more common than its counterpart Anticipatory H-shifting/spreading leftward (Hyman & Schuh 1974; Hyman 1978, 2017). This complies with the tendency for tone targets to be realized late, part of the literature on 'pitch' or 'peak delay' (Yip 2002; Kingston 2003; Akinlabi & Liberman 2007; Zhang & Meng 2016:172). Still, there are enough instances of leftward shift cross-linguistically that inferring its appearance in a hypothetical intermediate reconstruction is certainly plausible (see examples catalogued in Hyman 2017 and subsequent discussion).

Thus, both rightward and leftward shift took place in the development of Edoid, even resulting in 'diachronic oscillation': Proto-Edoid melodies **L-H and *L-HL become P-NC *H-L, which then become L-H in present-day Edo (i.e. **L-H > *H-L > L-H). This is in fact directly parallel to the development of Akan dialects as detailed in Stewart (1983). In Akan [aka], an original leftward shift of high tones affected the language as a whole (i.e. a "synchronic leftward tone-shift rule was already in operation as a synchronic rule in proto-Akan" – p. 240), and this rule was "subsequently reversed" only in the Asante dialect by a rightward shift. Under this paper's proposal, such a diachronic oscillation would have also taken place in the North-Central Edoid area.

Moreover, operations affecting and involving H tones (such as H-Lowering and H-Obligatoriness) are more common than those with L tones (such as L-Raising). H-Lowering operations are diverse, affecting H as a whole (e.g. *H > M in Ghotuo [aaa]), and H after both Hs (e.g. *H-H > H-L in Eruwa [erh]) and after Ls (e.g. PE **L-H > *L-L in P-DE). As are H-

Obligatoriness constraints, adding a H to either the left or right edge of the word depending on the language. In contrast, L-Raising essentially took place only twice: *H-L > L-HM (after shift) in Epie-Atissa [epi] (assuming an intermediate stage L-HL), and *H-L > H-ꜜH in Urhobo [urh] (and in this latter case, not without caveats – see §4.2). Still, there are certain changes which involve both L and H at the same time, as in the case of Rising-Markedness which changes *L-LH to L-L or L-H in order to avoid this kind of stem-level rise after a prefix L.

6.2 On the regularity of tone change

Our efforts to reconstruct proto-Edoid tone shows both the success and limits of regular sound change as applied to tone. The regularity of sound change principle (or 'Neogrammarian Hypothesis' – Crowley & Bowern 2010:170) states that sound change applies whenever the context is met (e.g. k → c / __i), which entails that all else being equal a sound change will affect all relevant words in the lexicon (for a recent summary and numerous references, see Labov 2020).

At the same time, there have been several challenges to this principle. Most relevant with respect to this paper, tone in particular shows frequent non-compliance with the regularity principle. In-depth surveys purporting irregularity in tone change go back at least to Cheng & Wang (1977), who document a lexical split of reflexes of Middle Chinese tone III in the Chinese variety Chaozhou [nan] – half into 'tone 2b' and half into 'tone 3b' – without a clear phonetic or phonological condition (see also Labov 1981). More recently, Dockum (2019:89ff.) summarizes the skepticism towards tone within the historical linguistics literature. Even in highlighting the success of the 'Tonal Comparative Method', Dockum ties regular tonal changes directly to segmental origins causing tone splitting, which implies that without clear tonogenesis events (as in the case of Edoid and most of Africa – §1) regularity in tone change is obscured.[12] As a whole, because the realization of tone as phonetic pitch is entirely relative, tone appears ripe for the generation of more surface variants which over time may result in daughter languages with more irregular tone changes compared to segmental changes.[13]

Within Edoid tone, we have posited subsets for virtually every correspondence set, which could be interpreted as irregular change. In most cases, only a small number of cognates deviate from the expected reflex, e.g. in P-DE *L-L remains L-L in Engenni [`enn`] in 83/86 instances. Only a small number have un unexpected high tone, namely the three cognates HAND_ARM [**ɔ́ɓɔ̀**], LEAF [**ɛ̀ɓí**], and VEHICLE_BOAT [**ɔ̀kɔ́**].

More serious are cases where the number of exceptional cognates is much higher. Examples include P-SW *L-L which are evenly split in Eruwa: roughly half remaining L-L (e.g. WORD_2 [**ɛ̀mì**] and CHARM_DRUG_3 [**ùmù**]) and just over half becoming [L-H] (e.g. WATER [**àmí**] and WAR_FIGHT_2 [**ɛ̀mɔ̋**]). In parallel, P-NC *L-L regularly remains L-L in Edo (e.g. TIME [**ɛ̀ɣɛ̀**]), but gains a H tone in approximately 20% of cognates (e.g. CHAIN [**éɣɛ̏**]).

Unexpected outcomes are also recurrent in the proto-melodies involving with H tones. For example, PE **H-H remains *H-H in most P-DE cognates but becomes *H-L about half as many times (cf. Degema SHEEP_1 [**óꜜsómáɲ**] < P-DE *H-H vs. NOSE [**ísúβèɲ**] < P-DE *H-L, both

[12] For example, Dockum (2019:100) writes: "the phonetic tones are not the immediate object of comparison in the Tonal Comparative Method. Rather, it is the historical tone categories and their conditioning environments that must be identified through uncovering their segmental origins. Thus, before there can be tonal correspondence sets, there must be segmental correspondence sets."

[13] At the same time, there are researchers whose painstaking work demonstrate not only that tone is largely regular, but that it does not change at faster or slower rates compared to segments (Auderset 2024, on the Mixtec family). At this point, it would be immature to compare rates of change for tones vs. segments in Edoid.

reconstructed as PE **H-H). And in Urhobo, we have said that the reflex of P-SW *H-L has two outcomes with an unclear conditioning environment: it is either retained as H-L (*n*=42) or becomes H-ꜜH with H-Raising (*n*=27).

Such cases of apparent exceptions to the regularity of sound change principle are the noise through which we observe systematic tone change. There are two opposing positions one could take in response to this 'noise'. One would be that irregular change is an inevitable conclusion, which would then reverse the burden of explanation to asking why certain changes are so overwhelming regular (e.g. P-DE *L-L becomes surface L-H in virtually all Degema cognates). The second would be to assume regular change in nearly all cases, with apparent irregularities due to hidden phonological environments (whether segmental or tonal), morphological factors (e.g. analogy), or kinds of dialect mixing (in line with the original Neogrammarian Hypothesis). We can zoom in on evaluating these opposing positions for Edoid within the South-West branch where it appears most pressing, which we turn to presently.

6.3 Is South-West Edoid conservative or innovative?

Related to the regularity of sound change is when to posit new tonal correspondences versus when to posit a single correspondence with a conditioning factor causing a split within that category. Nowhere is this more apparent than in SW Edoid. In §5.2 above, we discussed the 'high/low' grouping of Proto-Edoid, where some branches have an initial L- but others H-. We repeat this grouping in (60) below, which involve PE classes **L-H (PE-03), **L-HL (PE-04) and **H-L (PE-05). As can be seen, while DE, NC, and NW branches are basically uniform, SW shows three distinct reflexes (none of which are identical to the PE form). The question is whether this reflects tone class conservation in SW and merger in DE, NC, and NW, or is this class innovation strictly bound to SW? This has serious consequences to the inventory of tone changes in Edoid and its implications for the wider typology of tone change (§6.1).

(60) SW-Edoid: Tone class conservation or innovation?

Proto-Edoid			Proto-DE	**Proto-SW**	Proto-NC	Proto-NW
PE-03	**L-H	>	*L-L	***L-LH**	*H-L	*H-L
PE-04	**L-HL	>	(?)	***L-L**	*H-L	*H-L
PE-05	**H-L	>	*L-L	***L-H**	*H-L	*H-L

While our reconstructions assume more rather than fewer tone melodies which then underwent merger, let us entertain the alternative: the three-way contrast in P-SW in (60) is a split of a PE category. This innovation alternative may be plausible based on the geographic location of SW languages. The two largest SW languages – Urhobo and Isoko – exist in the forest/swamp transition zone of southern Nigeria and consist of expansive networks of speakers across a relatively large area. These societies show affinities and relations to the other larger kingdoms and populations in the area, such as Yoruba and Edo polities to the north/west, Igbo ones to the east, and Ijoid ones to the south (maps in (1) and (3)), and their position as a 'crossroads' for various groups may have resulted in innovation in tonal patterning. Indeed, ATR vowel harmony and the morphological noun class systems of Urhobo and Isoko are eroded (an innovation), especially compared to the much more fractured but conservative DE and NW branches at the geographic periphery of Edoid.

If SW is innovative, what were the (hidden) phonological or morphological environments which conditioned one tone group splitting into several? Unfortunately, no clear conditioning factor is apparent to which we could attribute the SW tone differences. There is no clear phonological factor – e.g. a consonantal feature which conditioned a tone change and then was subsequently lost, such as the proto-Edoid fortis/lenis distinction (Maddieson 1974) – nor a clear morphological factor – e.g. a prefix which appears only in SW words which exceptionally have a

H or L tone where we would otherwise expect the opposite.[14] Perhaps grammatical tone in compounds/possessives played a role in tone class splitting – whereby one or both of the nouns bear special tone properties as the partial or sole exponent of this construction – but presumably such effects would've been equally possible to result in splits in other Edoid branches.

The most promising direction involves syllable count. The stems of most Edoid languages are 1-2 syllables, but in past stages of the language this could've been 2-3. In fact, Elugbe's (1989a) segmental reconstructions frequently posit two stem vowels/syllables which have reduced in most modern languages, especially SW-Edoid. Representative examples include bisyllabic PE stems NIGHT_1 ****A-cʊNa**, CHIN ****A-gbhamhı**, FISH_1 ****E-chiənhi**, EAR ****ghU-chɔGi**, and SONG_1 ****I-yodho**, whose stems are all monosyllabic in the modern day SW language Urhobo (respectively, [**à-sɔ̃̀**], [**é-gbã̄**], [**è-rĩ̀**], [**ó-r̥ɔ́**], and [**ù-lè**]). Different waves of segmental reduction in the history of Edoid could have resulted in mismatches between tone and syllable count, leading to various kinds of distinct surface tone patterns from the same original tone class.[15]

As it stands, we require further data and scrutiny in order to adjudicate whether SW Edoid is conservative (entailing more PE tone classes) or innovative (entailing few classes). We posit more classes at the present moment as a hypothesis to be tested, which could later be collapsed if it were found to be unsupported.

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[14] By way of illustration, in the Ibilo dialect of Okpamheri (Elugbe 1973:257), certain nouns in the singular contain a singularizing prefix /**írl**-/ with an inherent high tone (compare plural **ìtò** 'buttocks' to its singularized form **írl-ètò**). It is plausible that some kind of comparable prefix could have existed in SW at some stage, and subsequently deleted leaving only a tonal trace.

[15] For similar ideas regarding apparent irregularities in tone correspondence and the role of segmental/syllable loss, see Beam de Azcona (2007:§2) on Southern Zapotec.

6.4 Implications for Proto-Benue-Congo (and Proto-Niger-Congo)

If the Proto-Edoid reconstructions prove successful (both the number of correspondence sets and the proposed melodies), what implications are there for Proto-Benue-Congo and ultimately Proto-Niger-Congo? As stated in §2, Edoid is but one of several parallel branches of Benue-Congo (detailed in the Map in (1)). Other major groups include Yoruboid, Igboid, Delta Cross, Bantoid (which includes Bantu), Nupoid, among several others. There exist already individual proposals reconstructing their proto-languages many with tone included, such as Proto-Yoruboid (Akinkugbe 1978; Pozdniakov 2018), Proto-Igboid (Williamson, Blench, & Ohiri-Aniche 2016), Proto-Lower Cross (Connell 1991) and Proto-Upper Cross (Dimmendaal 1978) within Delta Cross, and Proto-Grassfields within Bantoid (Elias, Leroy, & Voorhoeve 1984). How do the Edoid patterns line up these reconstructed forms?

At this point, no definitive correspondences emerge comparing the eight Proto-Edoid sets to these five ancestral languages, shown in the tables below. For comparative purposes, we also include in these tables two modern languages closely related to Edoid within Benue-Congo which are of unclear affiliation, Akpes (Lau 2021) and Ikaan (Salffner 2009). Starting with Proto-Edoid **L-L nouns (PE-01), a list of plausible cognates are in (61) (though note cognacy has not been verified for this exercise). PE **L-L corresponds to distinct L-L or M-M patterns in Akpes, and to distinct L, M, or H stems in Proto-Yoruboid with either a L or M on the initial vowel prefix. Similar variation exists for the other proto-languages, as shown.

(61) Proto-Edoid comparative data for **L-L (PE-01) nouns

	CHIN PE: ****à-gbhàmhì** **L-L (PE-01)	WATER PE: ****à-mìN** **L-L (PE-01)	NECK_THROAT_1 PE: ****ù-thòdhù** **L-L (PE-01)	TAIL_1 PE: ****ù-thìəmhì** **L-L (PE-01)	SALIVA PE: ****à-cràNì** **L-L (PE-01)	HEART PE: ****ù-ɗìə** **L-L (PE-01)
a. Akpes	**-**	**ī-ɲī**	**ɔ̄-hɔ̃̄**	**-**	**ì-sʷàj**	**-**
Ikaan	**ì-gbàgbénà**	**ù-mɔ̃́**	**-**	**-**	**-**	**-**
b. Proto-Yoruboid	***à-gbã̂**	***ō-mĩ̄**	***ɔ̄-rɔ̃̀**	***ù-rũ̀**	***ī-tɔ̃́**	***ɛ̀-dɔ̃̀**
Proto-Igboid	***à-`-gbã̀**	***m̀-mí-`-ɗí**	***ó-`-lúCĩ̀**	***ɔ́-`-dɔ̃̀**	***ɛ́-`-tɪ̃́ã́**	***ì-mé-tĩ̂**
Proto-Upper Cross	***-gbà**	***mà-ní**	**-**	**-**	***-tàj**	***-tímà**
Proto-Lower Cross	***m̀-báŋ**	***´-mɔ́ːŋ**	***ù-tɔ́ŋ**	***ù-sìm**	***é-táp**	***é-cìt**
Proto-Grassfields	**-**	***-mò´**	***-tóŋ´**	***-sáŋ´**	***-tué`**	***-tím`**

A comparable lack of systematic correspondence is seen with the other PE-sets, such as **L-H (PE-03) and **H-H (PE-08), in (62) and (63) respectively. In both these tables, the PE form corresponds to a range of tone patterns within an individual language, without a clear majority.

(62) Proto-Edoid comparative data for **L-H (PE-03) nouns

	ANIMAL_MEAT PE: ****ɛ̀-nhámhí** **L-H (PE-03)	HAIR PE: ****ʊ̀-tʊ́N** **L-H (PE-03)	HAND_ARM PE: ****ghʊ̀-ɓɔ́** **L-H (PE-03)	TONGUE PE: ****ʊ̀-dhámhí** **L-H (PE-03)	MOUTH PE: ****è-nuə́** **L-H (PE-03)	ELEPHANT PE: ****è-ní** **L-H (PE-03)
a. Akpes	**ɛ̀-nɔ̃́m**	**è-wùr**	**-**	**-**	**ō-nū**	**-**
Ikaan	**ɛ̄-nām**	**è-tìf**	**ɔ̄-bɔ̄**	**-**	**-**	**ěː-ní**
b. Proto-Yoruboid	***ɛ̄-rĩ̄ã̄**	***ō-rɔ̃̄**	***ɔ̄-b'ɔ́**	***è-dè**	***ā-rɔ̃̄ã̄**	***ē-rĩ̄**
Proto-Igboid	***á-`-nV̂C̃ɔ̃̀**	***ɗʊ́-`-gɪ̃́ɔ̃̀**	***ɔ́-bʊ̀-bà**	***ì-ɗí-`-ɗéCò**	***ɔ́-`-nɔ̃̌**	***é-`-nɪ̃́Nɪ̃̀**
Proto-Upper Cross	**-**	**-**	***-bɔ́k**	***-dák**	***-mà**	***-nì**
Proto-Lower Cross	***ú-nàm**	***í-dɛ̀t**	***ú-bɔ́k**	***ɛ́-lɛ́mɛ̀**	***í-núà**	***é-nìːn**
Proto-Grassfields	***-nàm`**	**-**	***-bó`**	***-lím`**	**-**	**-**

(63) Proto-Edoid comparative data for **H-H (PE-08) nouns

		EAR PE: **ghʊ́-chɔ́Gí **H-H (PE-08)	TREE PE: **ʊ́-tháNí **H-H (PE-08)	EGG PE: **dhí-kíNə **H-H (PE-08)	NOSE PE: **ɪ́-chuəNí **H-H (PE-08)	VEIN PE: **ú-niə́ **H-H (PE-08)	FLY_1 PE: **-ə́-khíNə **H-H (PE-08)
a.	Akpes	**ā-sūg**	**ɔ̄-hūnɛ̄**	**-**	**ā-hūǹ**	**-**	**ī-ʃī**
	Ikaan	**ʊ̀-r̥ʊ̀g**	**ò-hún**	**ì-ʃɛ̀**	**ɔ̀-kɔ̋rɔ̀m**	**-**	**-**
b.	Proto-Yoruboid	***ē-tí̋**	***ì-tì**	***ɛ̄-g’ī̄**	***ī-ŋmő**	**-**	***V̄-cī̃cī̃**
	Proto-Igboid	***ī́-ǹ-ƭɛ̋ì̄**	***ú-tî-tî**	***ì̄-kʷâ̄**	***é-`-mî**	**-**	***ó-gì-ì̄-gí̄**
	Proto-Upper Cross	***-ttóŋ(ì)**	***-tté**	***-kkèŋí**	***-djòná**	***-ní**	**-**
	Proto-Lower Cross	***ú-tɔ́ŋ**	***é-tíé**	***ŋ́-kíɛ̀n**	**-**	***ɔ́-lɔ̀ŋ**	***ú-sùŋ**
	Proto-Grassfields	***-túŋ´**	***-tí´**	**-**	***-duî**	**-**	***-dʒì**

The lack of any clear signal in these comparative data is sobering for the prospects of reconstructing tone in Proto-Benue-Congo, let alone a higher order phylum like Proto-Niger-Congo itself (cf. Hyman 2020 postulating *H and *L in Proto-Niger-Congo but without clear correspondence sets).[16] Once more data has been collected for the current synchronic systems of Benue-Congo coupled with systematic tonal reconstruction of the individual branches, comparison of Proto-Edoid with sister languages may be revisited.

7 Summary

The immediate goal of this paper was to reconstruct tone on nouns in the Edoid family of Nigeria, one branch within the Benue-Congo subfamily of the Niger-Congo phylum, the largest family in Africa. African languages are important to studies of tone diachrony as no original tonogenesis event is discernable, and we can thus assess how far back in time we can reconstruct tone using the comparative method without segmental correspondence. To this end, we developed the Proto-

[16] De Wolf (1971:51) posits *H and *L for Proto-Benue-Congo, but he too does not provide any correspondence sets.

Edoid Tone (PEdT) database drawn from 38 sources on 21 modern Edoid languages, totally over 3000 individual data points, from which 436 cognates can be identified. Based on the PEdT database, we systematically compared tone patterns across the four proposed branches of Edoid which led to a number of tonal correspondences in each branch reconstructed as combinations of tone on a noun class prefix and tone on a nominal stem (i.e. ‘nominal tone melodies’). Based on these intermediate reconstructions, we reconstructed tone in Proto-Edoid itself as consisting of the symmetrical set of melodies **L-L, **L-LH, **L-H, **L-HL, **H-L, **H-LH, **H-HL, and **H-H. From these Proto-Edoid melodies, we tracked several recurring tone changes in Edoid, the most common of which were Rightward H-Shift (e.g. the ‘Great Edo Tone Shift’ *H-L > L-H), H-Obligatoriness (i.e. adding a H to all-L words), and H-Lowering (i.e. *H > ꜜH, M, or L). These tone changes showed both the hallmarks of regular sound change as well as many unexpected reflexes which cannot be explained at this point, setting the stage for future work on Edoid tone especially as it relates to Proto-Benue-Congo and ultimately Proto-Niger-Congo tone.

8 References

Agheyisi, Rebecca N. 1986. *An Edo-English dictionary*. Enugu, Nigeria: Ethiope publishing corporation.

Agheyisi, Rebecca N. 1990. *A grammar of Edo.* UNESCO.

Aigbe, E. I. 1985. “Edo-English Dictionary”. Unknown publisher.

Akinkugbe, Olufẹmi Odutayo. 1978. A comparative phonology of Yoruba dialects, Iṣẹkiri and Igala. PhD thesis, University of Ibadan.

Akinlabi, Akinbiyi & Mark Liberman. 2007. Tonal complexes: The prosodic organization of tones. Rutgers Optimality Archive ROA: 911. https://roa.rutgers.edu/article/view/941.html

Auderset, Sandra. 2024. Rates of change and phylogenetic signal in Mixtec tone. *Language Dynamics and Change* 14, 1-41.

Aziza, Rose O. 1997. *Urhobo tone system*. PhD thesis, University of Ibadan.

Aziza, Rose O. 2003. Tonal alternations in the Urhobo noun phrase. *Studies in African Linguistics* 32(2). 1-24.

Aziza, Rose O. 2008. Neutralization of contrast in the vowel system of Urhobo. *Studies in African Linguistics* 37(1). 1-19.

Beam de Azcona, Rosemary G. 2013. Problems in Zapotec tone reconstruction. In Antić Zhenya, Charles B. Chang, Claire S. Sandy, & Maziar Toosarvandani (eds.), *Proceedings of the 33rd Annual Meeting of the Berkeley Linguistics Society: Special Session on Languages of Mexico and Central America*, 3-15. Berkeley: Berkeley Linguistics Society.

Blanc, Jean-François. 1986. Le verbe urhobo. PhD thesis, University of Grenoble.

Boyeldieu, Pascal. 2000. *Identité tonale et filiation des langues sara-bongo-baguirmiennes (Afrique centrale)*. Köln: Rüdiger Köppe Verlag.

Campbell, Eric. 2021. Why is tone change still poorly understood, and how might documentation of less-studied tone languages help? In Patience Epps, Danny Law, & Na’ama Pat-El (eds.), *Historical linguistics and endangered languages: Exploring diversity in language change*, 15–40. New York: Routledge.

Cheng, Chin-Chuan, & William S. Wang. 1977. Tone change in Chao-Zhou Chinese: A study in lexical diffusion. In W. S. Wang (ed.), *The lexicon in phonological change.* Berlin: De Gruyter. 86-100.

Connell, Bruce. 1991. Phonetic aspects of the Lower Cross languages and their implications for sound change. PhD thesis, University of Edinburgh.

Crowley, Terry & Claire Bowern. 2010. An Introduction to Historical Linguistics, 4th Edition. Oxford: OUP.

de Wolf, Paul P. 1971. *The noun class system of Proto-Benue-Congo*. The Hague: Mouton.

Dimmendaal, Gerrit J. 1978. The Consonants of Proto-Upper Cross and their Implications for the Classification of the Upper Cross Languages. Leiden: Leiden University.

Dockum, Rikker. 2019. The tonal comparative method: Tai tone in historical perspective. PhD thesis, Yale.

Donwa-Ifode, Shirley. 1986. Phonetic Variation in Consonants (Isoko). *Anthropological Linguistics* 28(2). 149-160.

Donwa-Ifode, Shirley. 1989. Prefix vowel reduction and loss of noun class distinction: The Edoid case. *Afrika und Obersee* 72. 229-253.

Egharevba, Jacob. 1968 [1934]. *A short history of Benin*. Ibadan: Ibadan University Press.

Eigbedion, Genevieve. 1985. Negation in Esan. MA Thesis, University of Ibadan.

Ejele, P.E. 1986. Transitivity, tense and aspect in Esan. PhD thesis, University College London.

Ekiugbo, Philip Oghenesuowho & Christian Ugo Chukwunonye Ugorji. 2019. A Descriptive Phonology of the Vowel System of Uvwiẹ, *Linguistique et langues africaines* 5. 89-107.

Elderkin, E. D. 2008. Proto-Khoe tones in Western Kalahari. In S. Ermisch (ed.), *Khoisan Languages and Linguistics: Proceedings of the 2nd international symposium, January 8–12, 2006, Riezlern/Kleinwalsertal*. Cologne: Research in Khoisan Studies 22. 87–136.

Elias, Philip, Jacqueline Leroy & Jan Voorhoeve. 1984. Mbam-Nkam or Eastern Grassfields. *Afrika und Übersee* 67. 31-107.

Elimelech, Baruch. 1976. *A tonal grammar of Etsako*. WPP 35. Los Angeles: Department of Linguistics, UCLA.

Elugbe, Ben O. 1973. A comparative Edo phonology. PhD thesis, University of Ibadan.

Elugbe, Ben O. 1977. Some implications of low tone raising in southwestern Edoid. *Studies in African Linguistics* Supplement 7. 53-62.

Elugbe, Ben O. 1979. Some tentative historical inferences from comparative Edoid studies. *Kiabàrà* 2(1). 82-101.

Elugbe, Ben O. 1986. The analysis of falling tones in Ghotuo. In Koen Bogers, Harry van der Hulst, & Marten Mous (eds.), *The phonological representation of suprasegmentals*. Dordrecht: Foris. 51-62.

Elugbe, Ben O. 1989a. *Comparative Edoid: Phonology and lexicon*. Port Harcourt, Nigeria: University of Port Harcourt Press.

Elugbe, Ben O. 1989b. The Tone System of Ghotuo. *Cambridge papers in Phonetics and Experimental Linguistics* 4. 1-21.

Elugbe, Ben O. 1989c. Edoid. In John Bendor-Samuel (ed.), *The Niger-Congo languages: A classification and description of Africa's largest language family*. Lanham: University Press of America. 291-304.

Elugbe, Ben O. & Klaus Schubert. 1976. Noun classes and concord in Oloma. *JWAL* 1-2. 73-84.

Emuekpere-Masagbor, G. A, 1997. Preverbal subject markers in Ivie. PhD thesis, Sherbrooke.

Greenberg, Joseph H. 1948. The tonal system of proto-Bantu. *Word* 4. 196-208

Hammarström, Harald & Forkel, Robert & Haspelmath, Martin & Bank, Sebastian. 2024. Glottolog 5.0. Leipzig: Max Planck Institute for Evolutionary Anthropology.

https://doi.org/10.5281/zenodo.10804357 (Available online at http://glottolog.org, Accessed on 2024-04-10.)

Hyman, Larry M. 1974. The great Igbo tone shift. In E. Voeltz (ed.), *Proceedings of the Third Annual Conference on African Linguistics*. Bloomington: Indiana University Press. 111–126.

Hyman, Larry M. 1978. Historical tonology. In V. Fromkin (Ed.), *Tone: A Linguistic Survey*. New York: Academic Press. 257-269.

Hyman, Larry M. 2006. Word-prosodic typology. *Phonology* 23. 225-257.

Hyman, Larry M. 2017. Synchronic vs. diachronic naturalness: Hyman & Schuh (1974) revisited. *UC Berkeley PhonLab Annual Report* 13(1).

Hyman, Larry M. 2020. On Reconstructing Tone in Proto-Niger-Congo. In Valentin Vydrin & Anastasia Lyakhovich (eds.), *In the hot yellow Africa*, 175-191. St. Petersburg: Nestor-Istoria.

Hyman, Larry M. & Russell G. Schuh. 1974. Universals of tone rules: Evidence from West Africa. *Linguistic Inquiry* 5(1). 81-115.

Hyslop, Gwendolyn. 2022. Toward a typology of tonogenesis: Revising the model. *Australian Journal of Linguistics* 42. 275–299.

Ikoyo-Eweto, E. O. & Philip Oghenesuowho Ekiugbo. 2017. A phonetic analysis of Uvwiẹ vowels. *Journal of West African Languages* 44(2). 9-26.

Iweh, Ode. 1983. La phonologie et le système nominal de l'Urhobo. PhD thesis, Université Stendhal (Grenoble 3).

Kari, Ethelbert E. 2004. *A reference grammar of Degema*. Köln: Rüdiger Köppe Verlag.

Kari, Ethelbert E. 2008. *Degema-English dictionary with English index*. Tokyo: Research Institute for Languages and Cultures of Asia and Africa (ILCAA), Tokyo University of Foreign Studies.

Kingston, John. 2003. Mechanisms of tone reversal. In Shigeki Kaji (ed.), *Cross-linguistic studies of tonal phenomena: Historical development, phonetics of tone, and descriptive studies*. 57-120. Tokyo: Institute for the Study of Languages and Cultures.

Klomp, Paul. 1993. A sketch of the phonology and morphology of Esan. MA thesis, Rijksuniversiteit te Leiden.

Labov, William. 1981. Resolving the Neogrammarian Controversy. *Language* 57(2). 267-308.

Labov, William. 2020. The regularity of regular sound change. *Language* 96(1). 42-59.

Ladefoged, Peter. 1968. *A phonetic study of West African languages: An auditory-instrumental survey*. Cambridge: CUP.

Lau, Jonas. 2021. A Digital Reference Grammar of Abesabesi. PhD thesis, Universität zu Köln.

Leben, William. 1973. Suprasegmental Phonology. PhD thesis, MIT.

Legbeti, Agnes Temítọ́pẹ́. 2022. Tone and Aspects of Grammar in Ósósọ̀, Edo, Nigeria. PhD thesis, University of Ibadan.

Lewis, Ademola Anthony. 2013. “North Edoid relations and roots. PhD thesis, University of Ibadan.

Maddieson, Ian. 1974. A possible new cause of tone-splitting—Evidence from Cama, Yoruba, and other languages. *Studies in African Linguistics* Supplement 5. 205-221.

Mafeni, B. O. W. 1969. Isoko. In E. Dunstan (ed.), *Twelve Nigerian languages: A handbook on their sound systems for teachers of English*. London & New York: Longmans; Africana Publishing.

Masagbor, Richard. 1989. Noun class and concord in Ivie. *JWAL* 19(1). 75-86.

Moñino, Yves. 1995. *Le Proto-Gbaya: Essai de linguistique comparative historique sur vingt-et-une langues d'Afrique centrale*. Paris: Peeters Publishers.

Olaoluwa, Ewekeye. 2012. Aspects of the Phonology of Ososo: An Autosegmental approach. MA thesis, University of Ibadan MA thesis.

Osiruemu, Evarista O. 2005. Tone and Grammar in Esan. MA thesis, University of Ibadan.

Pozdniakov, Konstantin. 2018. *The numeral system of Proto-Niger-Congo: A step-by-step reconstruction*. Berlin: Language Science Press.

Rolle, Nicholas, & Ethelbert E. Kari. 2022. Tone and prosodic recursion in Degema nouns and noun phrases. *Journal of African Languages and Linguistics* 43(2). 249-283.

Rolle, Nicholas, Florian Lionnet, & Matt Faytak. 2020. Areal patterns in the vowel systems of the Macro-Sudan Belt. *Linguistic Typology* 24(1). 113-179.

Ryder, A. 1969. *Benin and the Europeans 1485-1897*. London: Longman.

Sæbø, Lilja Maria, Eitan Grossman, & Steven Moran. 2025. Tonogenesis: A diachronic typology. *Diachronica* 42(3-4). 451-478.

Salffner, Sophie. 2009. Tone in the phonology, lexicon and grammar of Ikaan. PhD thesis, University of London.

Schaefer, Ron & Francis Egbokhare. 2007. *Emai dictionary*. Koln: Rudiger Koppe Verlag.

Schuh, Russell G. 2017. A Chadic cornucopia. Oakland, California: California Digital Library.

Stewart, John. 1983. The Asante Twi tone shift. In I. R. Dihoff (ed.), *Current Approaches to African Linguistics* 1. Dordrecht: Foris. 235-244.

Thomas, Elaine. 1978. *Grammar Description of the Engenni Language*. Arlington: SIL.

Thomas, Elaine & Kay Williamson. 1967. "Delta Edoid wordlists: A new version of the publication". [2009 updated by Roger Blench (ed.); originally published as Occasional Publication No. 8, Institute of African Studies, University of Ibadan, 1967].

Ukere, Anthony Obakponovwe. 2005. "Urhobo-English dictionary". Unpublished manuscript. [Edited by Roger Blench]

Welmers, William E. 1969. Structural notes on Urhobo. *Journal of West African Languages* 1. 85-107.

Williamson, Kay. 1989. Benue-Congo Overview. In John Bendor-Samuel (ed.), *The Niger-Congo languages: A classification and description of Africa's largest language family*. Lanham: University Press of America. 247-274.

Williamson, Kay, Roger Blench, & Chinyere Ohiri-Aniche. 2016. "Comparative Igboid". Unpublished Manuscript. Cambridge.

Yip, Moira. 2002. *Tone*. Cambridge: Cambridge University Press.

Zhang, Jie, & Yuanliang Meng. 2016. Structure-dependent tone sandhi in real and nonce words in Shanghai Wu. *Journal of Phonetics* 54(1). 169-201.

9 Appendices

9.1 Appendix 1: Edoid consonant correspondences

Using the comparative method, Elugbe (1989a) reconstructs the following consonants in Proto-Edoid (at left in (64) below), and the systematic reflexes in the daughter languages. Those with more than one are denoted with slashes within the cell, those for which the segment has deleted are denoted as the empty set '∅', and those for which data is missing are denoted with a question mark '?'. Elugbe's transcriptions are replaced with IPA equivalents, but recall that <h> in his transcription indicates a lenis consonant which we retain below.

(64) Proto-Edoid consonant correspondences (as proposed by Elugbe 1989a)

	DE-Edoid			SW-Edoid					NC-Edoid					NW-Edoid				
Proto-Edoid	[deg]	[enn]	[epi]	[erh]	[iso]	[oke]	[urh]	[evh]	[bin]	[ema]	[ets]	[une]	[aaa]	[olm]	[opa]	[uha]	[ehu]	[uku]
**p	f	f	p	f	f	f	ɸ	?	f	f	f	f	f	f	f	f	f/ɸ	f
**b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	v	b	b
**t	t	t	t	t	t	t	t	t	t	t	t	t	t	h	h	h	s	t̪
**d	d	d	d	s	t̪	s	c	c	d	d	d	d	d	z	dz	z	z	d̪
**c	s	s	s	s	s	s	s	s	s	s	ts	ts	ts	s	s	s	s	ɹ̥
**ɟ	z	z/j	z	?	dʒ	ɟ	ʒ	ʒ	z	z	dz	dz	dz	?	ɟ	ɟ	ɟ	j
**k	k	k	k	k	k	k	k	k	k	k	k	k	k	k	k	k	k	k
**g	?	?	g	?	ɣ	?	?	?	g	g	g	g	g	?	g	g	g	g
**kp	kp	kp	kp	kp	βˠ	kp	kp	kp	kp	kp	kp	kp	kp	kp	kp	kw	kw	kw
**gb	gb	gb	gb	gb	βˠ	kp	xw	xw	gb	gb	gb	gh	gb	gb	gb	gb	gb	gb
**ph	f	f	f	v	v̈	ɸ	ɸ	?	h	h	f	f/h	f	f	?	f	f	f
**bh	ɓ	ɓ	ɓ	ʋ	ʋ	ʋ	ʋ	ʋ	w/ʋ	β	ʋ	ʋ	ʋ	?	bh	v	b	b
**th	t	t	t	r/t	r/t	r̥	r̥	r/t	r̥	r	r	ɹ̥	r/t	h	h	s	ɹ̥	ɹ̥
**dh	ɗ	ɗ	ɗ	ɹ	ɾ	ɾ	l	ɾ	ɻ/ɹ	∅	ɻ	ɻ	∅/ɻ	ɻ/ɹ	r	r	r	d
**ch	s	s	s	j	z	r̥	r̥	s	h	h	s	sh	h/ʒ	z	z̈	s	ɹ̥	ɹ̥
**kh	k	k	k	h	h	j/w	h	h	x	x	kh	kh	x	?	kh	k	k	k
**gh	w	w	w	w	ɣ	h	ɣ	g	ɣ	∅	ɣ	h	∅	∅	g	g	g	g
**kph	kp	kp	kp	kp	βˠ	kp	xw	kp	kp	xw	kph	kph	xw	?	kph	kw	kw	kw
**gbh	gb	gb	gb	gb	gb	gb	gb	gb	gb	gb	gb	gb	gb	gb	gb	gb	w	gb
**ɓ	ɓ	ɓ	ɓ	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b
**ɗ	ɗ	ɗ	ɗ	d	d	d	d	d	d	d	d	d	d	d	d	d	d	d
**m	m	m	m	m	m	m	m	m	m	m	m	m	m	m	m	m	m	m
**n	n	n	n	n	n	n	n	n	n	n	n	n	n	n	n	n	n	n
**mh	m	m	m	ʋ	m	m	m	m	ʋ̃	m	mh	mh	mh	m	mh	m	m	m
**nh	n	n	n	ɹ̃	ɾ	ɾ	r̃	r̃	ɹ̃	∅	ɻ	r̃	r̃	r̃	r̃	n	n	n
**f	f	f	f	f	f/h	f	f	f	h	h	f	f	f	f	f	f	f	f
**v	v	v	v	v	v	v	v	v	v	v	v	v	v	v	v	v	v	v
**l	?	l	l	l	l	l	?	?	l	l	n	n	l	?	?	r	?	?
**j	j	j	j	?	?	?	?	?	j	∅	j	j	j	?	ʃ	h	?	h
**w	?	w	?	?	?	?	?	?	?	∅	∅	jh	∅	?	w	w	w	w

9.2 Appendix 2: Cognates grouped into the eight Proto-Edoid tone melodies

Here, we divide the full inventory of cognates (*n*=436) into the eight Proto-Edoid tone melodies. We provide the unique cognate code, its confidence score (H = high, M = medium, and L = low), a rough gloss for the cognate, and a segmental mnemonic of the common Edoid sound for that cognate. Note that this mnemonic should not be taken as a reconstruction, but simply a shorthand for reference purposes.

(65) PE-01 **L-L (*n*=113)

Code	Conf.	Cognate	Mnem.	Code	Conf.	Cognate	Mnem.
PE_002	L	AFTERNOON	*ota*	PE_364	H	LIFE_1	*agbon*
PE_004	H	AGE_GRADE_1	*otu*	PE_369	M	LOUSE_1	*iru*
PE_608	H	ANT_TERMITE_2	*ikhakha*	PE_383	H	MARKET	*aki*
PE_011	M	ANT_TERMITE_4	*inwai*	PE_396	L	MONKEY_2	*ekhokho*
PE_017	H	ANUS	*unuso*	PE_708	M	MONKEY_3	*ekha*
PE_018	H	APE	*osia*	PE_399	H	MOON	*uki*
PE_023	M	ASH	*emhue*	PE_413	M	MUD_1	*ogodo*
PE_025	H	AXE_2	*ugama*	PE_424	H	NAVEL_1	*ukho*
PE_030	H	BAG_1	*akpa*	PE_427	H	NECK_THROAT_1	*othoru*
PE_040	M	BAT_1	*ogbom*	PE_432	M	NEEDLE_2	*udiding*
PE_483	M	BILE	*elo*	PE_435	M	NET	*aga*
PE_366	M	BOUNDARY_3	*usi*	PE_438	L	NEW_N	*ogbon*
PE_089	L	BUFFALO_1	*ede*	PE_451	L	PANGOLIN	*ekhui*
PE_091	H	BUFFALO_2	*adhoghi*	PE_485	L	PARLOUR_PALACE_2	*eguare*
PE_095	H	BUNDLE_1	*utunu*	PE_713	M	PENIS_2	*ekhia*
PE_098	H	BUSH_DOG	*aghwa*	PE_458	H	PERSON_3	*ogbo*
PE_100	H	BUSH_FOWL	*awai*	PE_457	M	PERSON_4	*ebholo*
PE_107	H	CALABASH_1	*akpa*	PE_465	M	PIG_1	*eci*
PE_105	M	CALABASH_4	*obere*	PE_467	L	PIG_3	*elede*
PE_120	L	CHAIN	*ighan*	PE_479	L	PROFIT	*ere*
PE_123	H	CHALK	*othue*	PE_487	H	RAT_2	*oji*
PE_127	H	CHARM_DRUG_3	*ukhumun*	PE_498	M	RING	*utosa*
PE_344	H	CHIEF_KING_1	*ovie*	PE_512	H	RUBBISH	*ghuku*
PE_345	H	CHIEF_KING_2	*ogie*	PE_513	M	SALIVA	*aselen*
PE_134	H	CHIN	*agban*	PE_515	M	SAND_1	*ubum*
PE_149	H	COCK	*okpa*	PE_517	M	SAND_3	*ekhae*
PE_169	M	CROCODILE_1	*ejere*	PE_537	H	SICK_N_3	*emiami*
PE_170	H	CROCODILE_2	*oni*	PE_538	H	SIDE	*uphenhe*
PE_171	H	CROCODILE_3	*eva*	PE_543	H	SLAVE_1	*oghunmun*
PE_176	M	CUTLASS_2	*ogidi*	PE_549	M	SMOKE_1	*ubhuri*
PE_191	M	DOOR_2	*egu*	PE_550	L	SMOKE_2	*ighogho*

Code	Conf.	Cognate	Mnem.	Code	Conf.	Cognate	Mnem.
PE_207	H	ELDER_3	*omadhen*	PE_557	H	SNAKE_3	*aka*
PE_214	H	EYE	*edho*	PE_558	M	SOAP_1	*osa*
PE_215	H	FAECES_1	*icon*	PE_560	H	SOIL_1	*eken*
PE_217	H	FAME	*uci*	PE_563	H	SOME	*ivu*
PE_222	H	FATHER_2	*etha*	PE_566	H	SONG_1	*iyodho*
PE_224	M	FEAR_2	*uzo*	PE_570	M	SOUP	*usomi*
PE_227	M	FIRE	*ethale*	PE_579	H	SPIRIT	*echi*
PE_228	H	FISH_1	*echeni*	PE_581	M	SQUIRREL	*atan*
PE_236	L	FOETUS	*akpa*	PE_597	M	SUN	*ovoni*
PE_238	H	FOOL	*ekpa*	PE_598	M	SWAMP	*ekhworo*
PE_240	M	FOOT_LEG_2	*owo*	PE_601	H	SWORD_1	*aberen*
PE_250	H	FRIEND_1	*oce*	PE_603	M	TAIL_1	*uthumu*
PE_274	H	GRASS_1	*eben*	PE_604	M	TAIL_2	*uyae*
PE_279	H	GRAVE	*udi*	PE_624	H	TIME	*eghidhe*
PE_283	H	GROUND	*oto*	PE_630	H	TOOTH	*akon*
PE_295	H	HAWK_2	*agodi*	PE_631	H	TOP	*ekhunhi*
PE_300	H	HEART	*udu*	PE_712	H	URHOBO_TRIBE	*uchobo*
PE_304	H	HILL_1	*ogbolo*	PE_663	H	WATER	*amin*
PE_310	H	HOE_3	*otho*	PE_664	M	WEAVER_BIRD	*akha*
PE_315	L	HOLE_4	*eye*	PE_669	M	WELL_N	*ochade*
PE_319	M	HORN_3	*ukphani*	PE_681	H	WOMAN_WIFE_2	*okpoco*
PE_325	H	HOUSE_2	*apen*	PE_685	H	WORD_2	*emholi*
PE_341	H	JUNCTION	*ada*	PE_687	M	WORK_1	*ogili*
PE_064	L	KOLANUT_2	*edun*	PE_688	L	WORK_2	*akanya*
PE_357	H	LAW	*uchi*	PE_693	M	YAM_2	*obhila*
PE_363	H	LEOPARD	*ekpen*				

(66) PE-02 **L-LH (n=27)

Code	Conf.	Cognate	Mnem.	Code	Conf.	Cognate	Mnem.
PE_028	H	BACK_1	*uke*	PE_377	M	MAN_2	*okpea*
PE_067	M	BLOOD_2	*izala*	PE_387	M	MAT_3	*ewaa*
PE_112	H	CAMP	*ago*	PE_431	M	NEEDLE_1	*egbede*
PE_165	L	COWIFE_1	*odhue*	PE_446	L	OLD_WOMAN	*edede*
PE_206	M	ELDER_4	*okpa*	PE_447	M	ORACLE	*epha*
PE_246	L	FOREST_5	*oholi*	PE_453	H	PARLOUR_PALACE_1	*ogua*
PE_251	H	FRIEND_2	*ugbali*	PE_480	M	PROVERB	*itan*
PE_305	M	HILL_2	*egeli*	PE_527	M	SHEEP_2	*ogode*
PE_311	M	HOLE_1	*udamu*	PE_545	H	SLAVE_2	*ovien*
PE_312	H	HOLE_2	*udhodho*	PE_576	M	SPEAR_1	*ogaga*
PE_314	L	HOLE_5	*uvun*	PE_577	H	SPIDER_2	*akpakpa*
PE_702	H	HORN_1	*ighodho*	PE_633	M	TORTOISE_2	*owoy*
PE_326	H	HOUSE_5	*odi*	PE_585	M	WALKING_STICK	*okpo*
PE_370	M	LOUSE_2	*uduv*				

(67) PE-03 **L-H (*n*=38)

Code	Conf.	Cognate	Mnem.	Code	Conf.	Cognate	Mnem.
PE_010	H	ANIMAL_MEAT	*elamen*	PE_340	M	JUJU_MAGIC	*ejo*
PE_012	H	ANT_TERMITE_3	*echiri*	PE_355	H	KOLANUT_1	*ebhele*
PE_059	M	BELLY_GUTS_2	*ovumu*	PE_360	H	LEAF	*ebe*
PE_071	H	BODY_2	*ogbe*	PE_412	H	MOUTH	*unu*
PE_084	M	BREAST	*enye*	PE_421	L	NAME_3	*ude*
PE_097	H	BUSH_CAT	*etha*	PE_425	M	NAVEL_2	*utule*
PE_155	H	CORPSE_1	*odhimhin*	PE_439	M	NIGHT_1	*ason*
PE_180	M	DAY	*ede*	PE_443	H	OIL	*ebhili*
PE_181	H	DEATH	*ughudhi*	PE_546	M	SLEEP	*ovise*
PE_182	H	DEBT	*oca*	PE_610	M	THIEF	*ozigha*
PE_187	H	DOCTOR	*obo*	PE_711	M	THING_2	*omhina*
PE_210	H	ELEPHANT	*eni*	PE_629	H	TONGUE	*elamhen*
PE_237	M	FOOD	*ebale*	PE_646	M	TWENTY	*iyow*
PE_264	L	GIRL_2	*ogboto*	PE_649	H	URINE_1	*aphien*
PE_288	H	HAIR	*eto*	PE_069	H	VEHICLE_BOAT	*oko*
PE_289	H	HAND_ARM	*obo*	PE_659	M	WAR_FIGHT_2	*iwumu*
PE_291	H	HAT	*athu*	PE_704	M	WIND_3	*avere*
PE_299	M	HEAD	*utomu*	PE_691	H	WORM_2	*okhori*
PE_322	M	HOUSE_1	*uwaa*	PE_699	H	YEAR	*ukpe*

(68) PE-04 **L-HL (*n*=13)

Code	Conf.	Cognate	Mnem.	Code	Conf.	Cognate	Mnem.
PE_024	M	AXE_1	*esuwe*	PE_428	H	NECK_THROAT_2	*akhole*
PE_096	M	BUNDLE_2	*ekuru*	PE_434	M	NEST	*ekoo*
PE_125	H	CHARM_DRUG_1	*eboo*	PE_449	H	PALM_1	*udi*
PE_141	M	CLOTH_1	*ukpon*	PE_514	M	SALT	*umhen*
PE_188	M	DOG_1	*abhua*	PE_567	H	SONG_2	*iviei*
PE_223	H	FEAR_1	*ophen*	PE_697	H	YAM_4	*emadhe*
PE_281	H	GREY_HAIR	*eda*				

(69) PE-05 **H-L (n=20)

Code	Conf.	Cognate	Mnem.	Code	Conf.	Cognate	Mnem.
PE_014	H	ANTELOPE_1	*ethue*	PE_621	M	NECK_THROAT_5	*ovo*
PE_086	M	BRIDE	*opha*	PE_459	M	PEPPER_1	*isibo*
PE_148	H	COAL	*ibi*	PE_680	H	PERSON_5	*okhuo*
PE_158	H	COTTON	*odhudhu*	PE_469	M	PLACE	*esi*
PE_267	M	GOAT_1	*ebhe*	PE_540	M	SKIN_1	*ophian*
PE_328	H	HUNTER	*ochue*	PE_559	M	SOAP_2	*okhue*
PE_386	H	MAT_2	*ore*	PE_593	M	STRANGER	*arara*
PE_394	M	MIST	*awili*	PE_632	M	TORTOISE_1	*egui*
PE_395	M	MONKEY_1	*enwen*	PE_655	M	VULTURE	*ugulu*
PE_415	M	MUSHROOM	*usun*	PE_684	H	WORD_1	*ota*

(70) PE-06 **H-LH (n=54)

Code	Conf.	Cognate	Mnem.	Code	Conf.	Cognate	Mnem.
PE_606	M	ANT_TERMITE_1	*edon*	PE_292	L	HAWK_1	*odegbe*
PE_015	M	ANTELOPE_2	*ujo*	PE_293	M	HAWK_4	*okpare*
PE_031	H	BAG_2	*akpo*	PE_327	M	HOUSE_4	*ugbehe*
PE_039	L	BASKET_1	*agban*	PE_339	H	JOKE	*oden*
PE_042	M	BAT_2	*akogan*	PE_352	L	KNIFE_3	*abee*
PE_048	L	BEAN_1	*iphie*	PE_375	H	MAIZE	*oka*
PE_049	H	BEAN_2	*ichilen*	PE_378	M	MAN_3	*odzawoli*
PE_054	H	BED	*ukpale*	PE_390	H	MESSENGER	*oko*
PE_056	M	BEE	*owon*	PE_410	H	MOTHER_2	*inene*
PE_063	H	BIRD	*apheni*	PE_414	M	MUD_2	*ebhete*
PE_078	H	BOUNDARY_1	*ughuru*	PE_444	H	OKRA	*ikhiabho*
PE_081	M	BOW	*uchenlen*	PE_502	M	ROAD_1	*uwe*
PE_087	M	BROOM	*uhwere*	PE_525	M	SHADOW	*aghoghon*
PE_174	L	CALABASH_3	*ukpu*	PE_548	L	SMELL	*ewia*
PE_126	H	CHARM_DRUG_2	*ibi*	PE_551	M	SNAIL	*urhue*
PE_703	H	COLD_N	*onini*	PE_586	M	STONE_1	*udo*
PE_162	H	COW	*emila*	PE_615	H	THIRTY	*ogban*
PE_168	M	CRICKET	*oche*	PE_618	H	THORN_2	*ocha*
PE_183	M	DEEP_WATER	*eho*	PE_626	H	TODAY	*elena*
PE_192	M	DOVE	*idu*	PE_628	L	TOMORROW	*akhwe*
PE_226	H	FENCE	*ogba*	PE_643	H	TROUBLE	*uko*
PE_229	H	FISH_2	*iphele*	PE_661	H	WAR_FIGHT_1	*okhoni*
PE_297	M	GOAT_4	*okiri*	PE_665	H	WEEK	*ujola*
PE_277	H	GRASSCUTTER_1	*udi*	PE_694	M	YAM_3	*ole*
PE_472	H	GRASSCUTTER_3	*evuen*	PE_696	M	YAM_5	*okpei*
PE_473	H	GRASSCUTTER_4	*origbe*	PE_153	M	YAM_6	*iyokho*
PE_278	H	GRASSLAND	*ato*	PE_700	H	YESTERDAY	*ude*

(71) PE-07 **H-HL (n=15)

Code	Conf.	Cognate	Mnem.	Code	Conf.	Cognate	Mnem.
PE_073	M	BONE_1	*igwe*	PE_376	H	MAN_5	*ochi*
PE_130	M	CHICKEN	*okhokho*	PE_484	H	RACE	*une*
PE_179	H	DARKNESS	*ubia*	PE_499	M	RIVER_1	*ede*
PE_185	M	DIRT	*inwa*	PE_509	H	ROPE_1	*udhin*
PE_190	M	DOOR_1	*ode*	PE_510	H	ROPE_2	*upi*
PE_232	M	FLESH	*ubhon*	PE_590	M	STORY_1	*okhai*
PE_261	M	GIFT	*eche*	PE_670	M	WOMAN_WIFE_5	*amhe*
PE_263	M	GIRL_1	*ote*				

(72) PE-08 **H-H (n=32)

Code	Conf.	Cognate	Mnem.	Code	Conf.	Cognate	Mnem.
PE_052	M	BEAN_3	*eda*	PE_373	L	LOUSE_3	*iyen*
PE_338	L	BELLY_GUTS_4	*ibie*	PE_404	M	MOSQUITO_2	*unwe*
PE_106	H	CALABASH_2	*uko*	PE_417	H	NAIL_1	*ephinlen*
PE_122	H	CHAIR	*agala*	PE_442	H	NOSE	*ichuen*
PE_131	H	CHILD_1	*omo*	PE_456	H	PENIS_1	*ekue*
PE_160	M	COUGH_N	*ohuoren*	PE_474	M	POT	*aki*
PE_115	H	COURT_CASE	*ejo*	PE_516	H	SAND_2	*ekpen*
PE_197	H	EAR	*echo*	PE_526	H	SHEEP_1	*ochuma*
PE_202	H	EGG	*eken*	PE_528	M	SHELL	*igho*
PE_220	L	FARM_2	*udu*	PE_582	M	STALL	*ogbha*
PE_234	M	FLY_1	*ikhien*	PE_642	H	TREE	*uthan*
PE_242	H	FOREST_1	*egbo*	PE_651	H	VEIN	*unhien*
PE_282	H	GRINDING_STONE	*olalo*	PE_636	M	VILLAGE_1	*oweey*
PE_336	H	INSECT	*ugo*	PE_656	M	WAIST	*ekun*
PE_343	M	KIDNEY	*iphuphu*	PE_675	M	WING	*uvu*
PE_365	L	LIFE_2	*aye*	PE_679	L	WITCH_WIZARD_2	*oso*